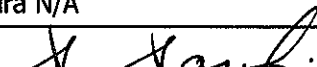
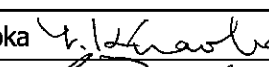
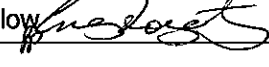
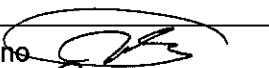


DUBAI RAPID LINK CONSORTIUM

DUBAI METRO PROJECT OFFICE

Contract No.: DM001		CDRL No.:	
Project Title: DUBAI METRO		CDRL Title:	
Document Title: Rashidiya Main Depot Structural Design Calculations Building 24 – Electrical Store DCP 3 Submission			
Revision History			
A4	16-May-07	Second issue	Signed below
A2	12-Nov-06	First Issue	Signed below
MARK	DATE	DESCRIPTION	RAIL
APPROVED		Project Director	S. Shimamura N/A
		Deputy Project Director	S. Sasaki 
		Checked By (Project Director)	Y. Hiraoka 
		Checked By (QA/QC Manager)	J. Whorlow 
Checked By	N/A	Checked By (Project Manager)	N/A
Checked By	N/A	Checked By (Engineering Manager)	M. Makino 
Prepared By	N/A	Issued By	R. Maney 
DATE	-	DATE	16 May 2007
RAIL SYSTEM		CIVIL JV	
CONTRACTOR'S DOCUMENT No.:		DOCUMENT No.:	REVISION
		DM001-E-MAO-CDP-DR-DCC-371952	A4

**Dubai Metro, UAE
Rashidiya Depot**

**Structural Design Calculation
(Enhanced Design)
Building 24 – Electrical Store**

May 2007

Client: JT Metro JV	Contract No. (if any):
Project Title: Dubai Metro, UAE	Project No.: JRD4561
Document No.: DM001-E-MAO-CDP-DR-DCC-371952	Controlled Copy No.:
Document Title: Main Depot- Rashidiya Structural Design Calculation (Enhanced Design) Building 24 – Electrical Store	
Covering Letter / Transmittal Ref. No.: To be advised	Date of Issue: 02 May 2007

Revision, Review and Approval/Authorization Records

Revision	Description	Prepared by / date	Reviewed by / date	App. or Auth. by / date
A4	Issued for Approval (ENHANCED DESIGN)	CK Lam May 07	P. Shepherd-Smith May 07	B. Maney May 07
A3	Issued for Approval	CK Lam Nov 06	P. Shepherd-Smith Nov 06	B. Maney Nov 06
A2	Issued for Approval	CK Lam Nov 06	P. Shepherd-Smith Nov 06	B. Maney Nov 06
A1	Internal Review	CK Lam Oct 06	P. Shepherd-Smith Oct 06	B. Maney Oct 06

Distribution (if insufficient space, please use separate paper)

Controlled Copy No.	Issued to
1	JT Metro JV
2	Atkins (Office Copy)
3	Atkins (P. Shepherd-Smith)
4	Meinhardt Infrastructure (CK Lam)

Note: App. and Auth. mean "Approved" and "Authorized" respectively.

Document No.: DM001-E-MAO-CDP-DR-DCC-371952

Rev A4

Date: 02 May 2007

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2	FOUNDATION DESIGN CHECK	032
3	CONNECTIONS DESIGN	037

1 TYPICAL PORTAL FRAME TRANSVERSE ANALYSIS

Document No.: **DM001-E-MAO-CDP-DR-DCC-371952**

Rev A4

Date: 02 May 2007

Page



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Job No

Sheet No

1

Rev

Part

Job Title

Ref

By

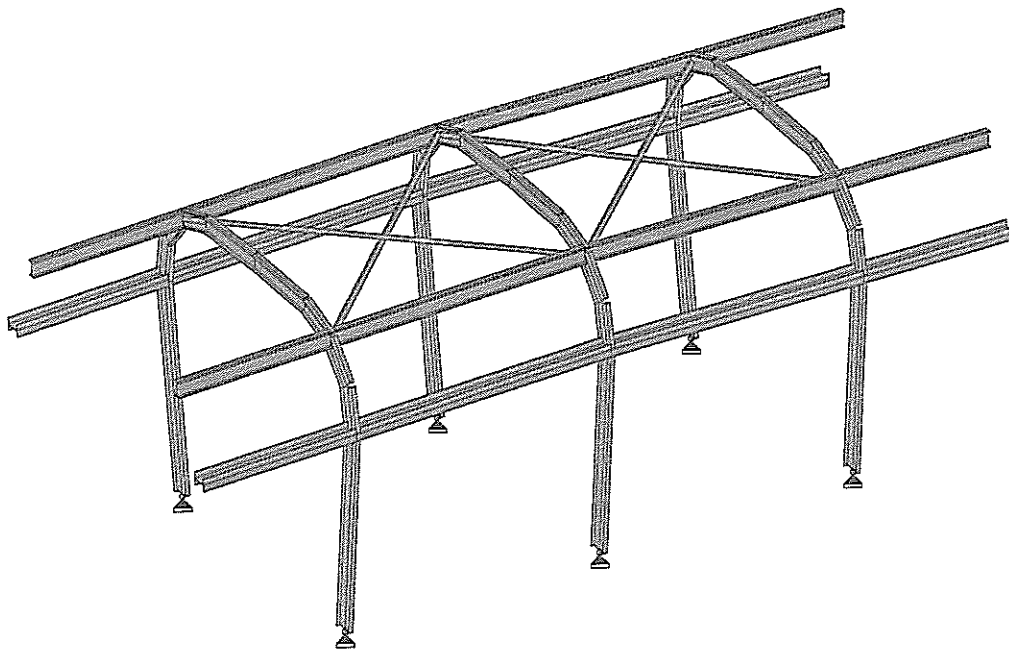
Date April 2007

Chd

Client

File Electrical Store.std

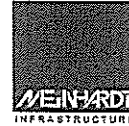
Date/Time 28-Apr-2007 15:18



Load 1

ELECTRICAL STORE

(RASHIDIYA - BUILDING 24)



Site:	Dubai Metro Project	Date:	27-Apr-07	Job No.:	68010.01
	Electrical Store				
	Loadings	Designed by: Rudy		Sheet No.:	

Roof Loadings :**DL** W_{DL} :

Sheeting	0.1 kPa		
Purlins	0.1 kPa		
Insulation	0.3 kPa		
M&E Finishes	0.3 kPa		
	<u>0.8 kPa</u>	x 1 =	0.8 kN/m

LL

0.6 kPa	x 1 =	0.6 kN/m
---------	-------	----------

P_{LL} : A/C equipment S/W
 = 5 kN

(point load at mid-span of bay 2, 4)

Notional horizontal load, Ph
 (Load Case of 1.4DL + 1.6LL)

Tributary load area for the roof
 = $(10+2*3)*(5+2*1.5)$
 = 128 m²

Total Dead Load
 102.4 kN

Total Live Load
 81.8 kN
 (Including A/C equipment)

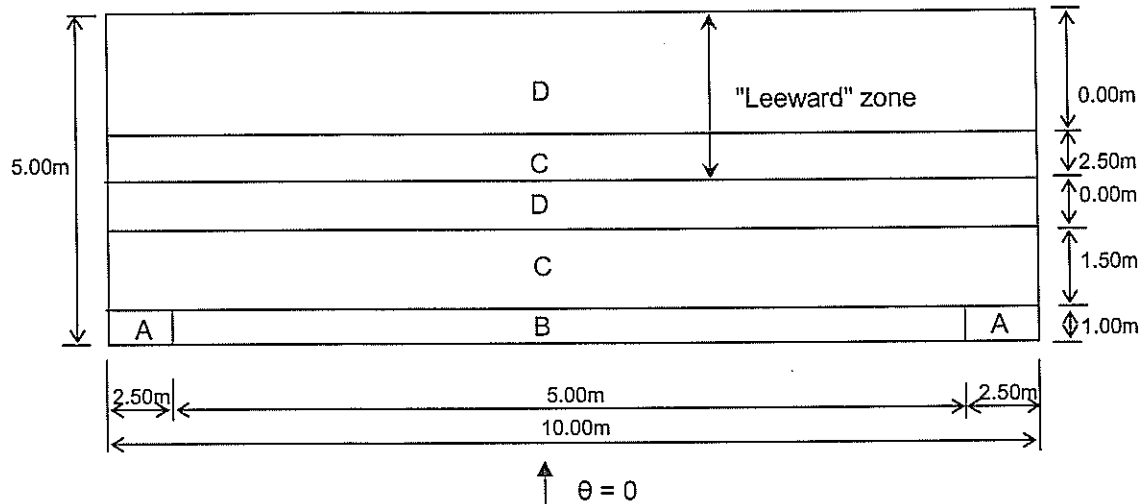
Tributary load area for the wall
 = $6*2*(10+2*3+5+2*1.5)$
 = 288 m²

Total Dead Load
 144 kN
 (excluding the M&E Finishes)

Total Live Load



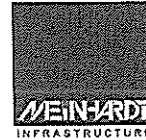
Site:	Dubai Metro Project	Date:	27-Apr-07	Job No.:	68010.01
	Electrical Store				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to eave)
PLAN


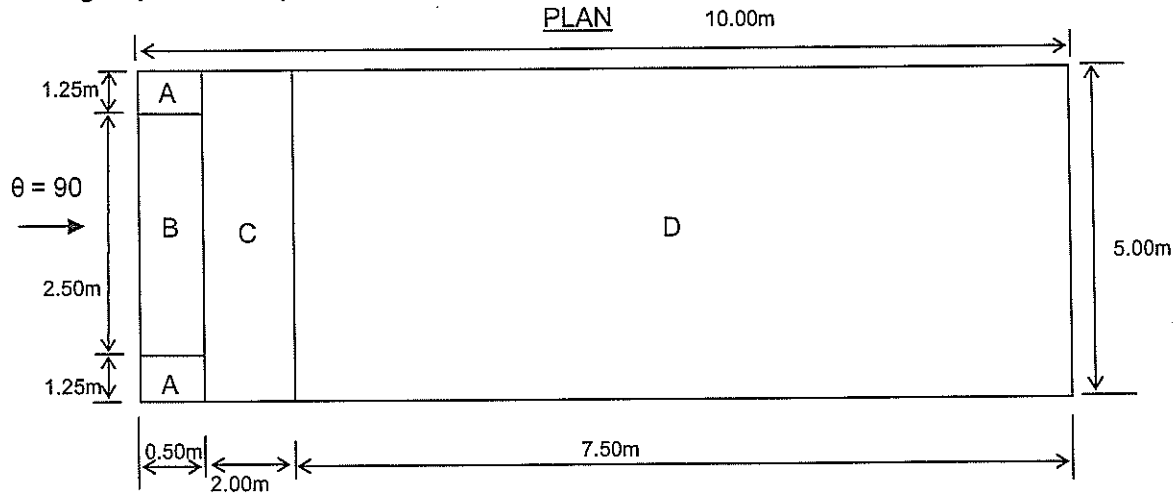
Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 10 \text{ m} \\
 H &= 6 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 10 \\
 &= 10 \text{ m}
 \end{aligned}$$

Zone	Windward		Leeward		
	Area (m ²)	Cpe	Area	Cpe1	Cpe2
A	5.0	-2			
B	5.0	-1.4			
C	15.0	-0.7	25.0	-0.7	-0.7
D	0.0	-0.2	0.0	-0.2	0.2
Total	25.0		25.0		
Average		-1.10		-0.70	-0.70



Site:	Dubai Metro Project	Date:	27-Apr-07	Job No.:	68010.01
	Electrical Store				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Average Cpe for roof (Wind direction normal to gable end)


Design as flat roof (with sharp eaves)

$$\begin{aligned}
 B &= 5 \text{ m} \\
 H &= 6 \text{ m} \\
 b &= \text{smaller of } (B, 2H) \\
 &= 1 \times 5 \\
 &= 5 \text{ m}
 \end{aligned}$$

For maximum uplift, assume zone C coefficient =>

$$C_{pe} = -0.7$$

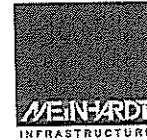
For maximum downward force, assume zone D coefficient =>

$$C_{pe} = 0.2$$

Wind Load Computation

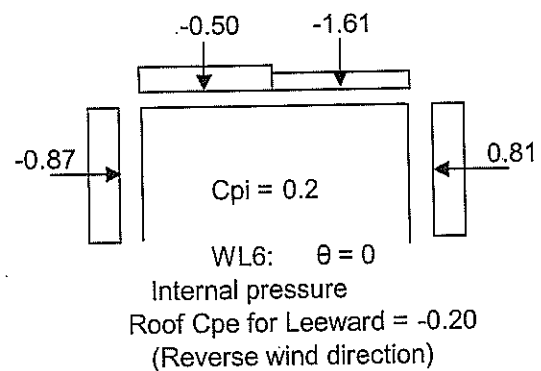
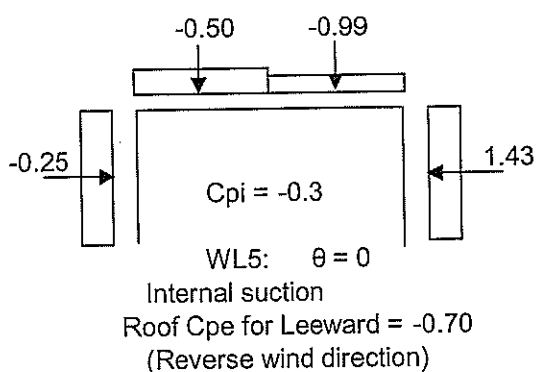
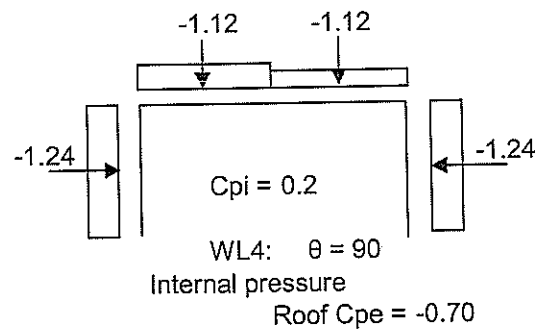
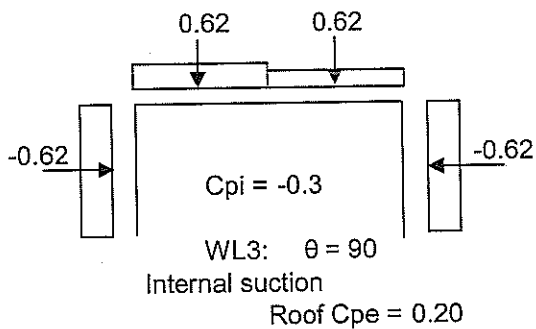
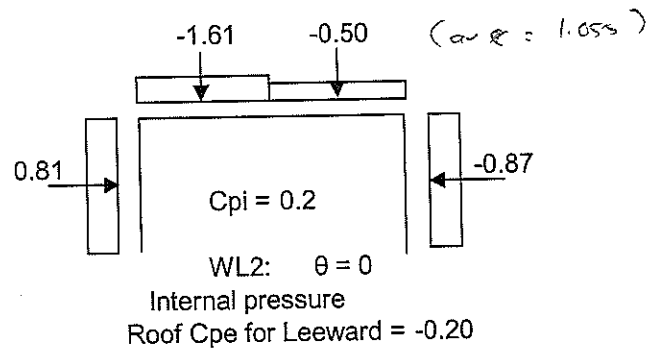
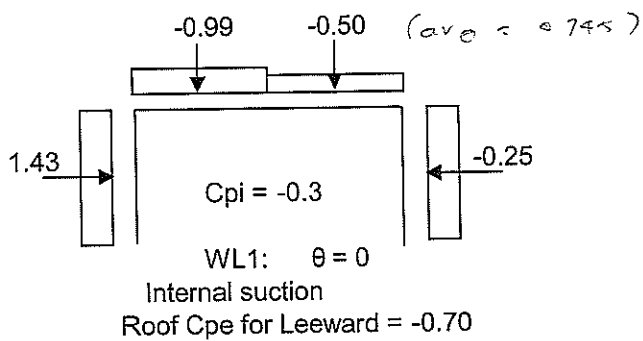
$$\begin{aligned}
 D &= 5.00 \text{ m} \\
 H &= 6 \text{ m} \\
 D/H &= 0.83
 \end{aligned}$$

Elements	Effective Wind Speed V_e m/s	Dynamic Wind Pressure q kN/m ²	External Pressure Coefficient, C_{pe}			
			$\theta = 0$ (normal to eave)		$\theta = 90$ (normal to gable end)	
			Windward	Leeward		
Roof	45	1.24	-1.10	-0.70	-0.70	-0.7 (Zone C) 0.2 (Zone D)
Wall	45	1.24	0.85	-0.5	-0.5	-0.8 (Zone B) -0.8 (Zone B)



Site:	Dubai Metro Project	Date:	27-Apr-07	Job No.:	68010.01
	Electrical Store				
	Wind Load Computation	Designed by:	Rudy	Sheet No.:	

Wind Load = 1 x q x (Cpe - Cpi) kN/m (For unit width)



Site: Dubai Metro Project
Electrical Store
Earthquake Loading

Date: 27-Apr-07

Job No.: 68010.01

Designed by: Rudy

Sheet No.:

Estimation of Earthquake Loading

Total Dead Load on the Building Column W = Given below (including part of Live Load)
Height of the Column h_n = Given below

Design Information:

- 1) Seismic Zone = 2A (from Design Brief)
2) Occupancy Category = 1 (Table- 16-K, Conservatively adopt 1)
3) Lateral Force procedure = **Static Procedure** (Section 1629.8 seismic zone 2A, structures not more than 5 stories)

Earthquake Loads

- 4) **Base Shear** $V_{base-S} = (C_v \times I \times W) / (R \times T)$ (Section 1630.2.1)
Seismic Coefficient $C_v = 0.32$ (Table 16-R assuming soil profile type S_D)
Importance Factor $I = 1.00$ (Table 16-K)
Numeric. Coef: Strength & Ductility $R = 4.50$ (Table 16-N for OMRT steel only)
Elastic Fundamental Period of Vibration $T = C_t (h_n)^{3/4}$ Sec.
Numerical Coefficient $C_t = 0.0853$ (Steel frame t.b.c)

- 5) **Limiting Values of Base shear** $V_{base-S-Max} = (2.5 \times C_a \times I \times W) / R = 35.11$
 $V_{base-S-Min} = 0.11 \times C_a \times I \times W = 6.95$

Design Earthquake Loads

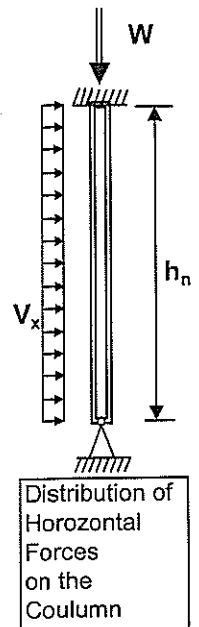
(Section 1630.6)

- $E_x = \rho \times \text{Min. } (E_h \text{ or } V_{base-S-Max}) + E_v$
 $E_{x-max} = \Omega \times \text{Min. } (E_h \text{ or } V_{base-S-Max})$
 $V_{base-S} = E_h =$ Earthquake load in the element due to the base shear, V , as set out above (for non-structural components refer to section 1632)
Vert. component of EQ ground motion $E_v = 0.5 C_a I D$ (Table 16-K)
Seismic Coefficient $C_a = 0.22$ (Table 16-Q, for soil type S_D)
Seismic Force Amplification Factor $\Omega = 2.80$ (Table 16-N)
Dead Load on the element $W = D =$ Given below
 $\rho = 2 - 6.1 / (r_{max} \sqrt{A_B}) = 1.25$ (conservative) ($1.0 \leq \rho \leq 1.5$)
 $r_{max} =$ The maximum element-storey shear ratio
 $A_B =$ Ground floor area of the structure

Distribution of Horizontal Forces:Distribution of Horizontal Force on the Column $V_x = E_x / h_n$

Col	Gr. Load, W , kN	Ht h_n , m	Time, T sec.	V_{base-S} kN	V_{base-S} kN	E_v kN	E_x kN	E_{x-max} kN	V_x kN/m	V_{x-max} kN/m
C1	287	6.00	0.33	62.5	35.11	31.603	75.5	98.3	12.6	16.4

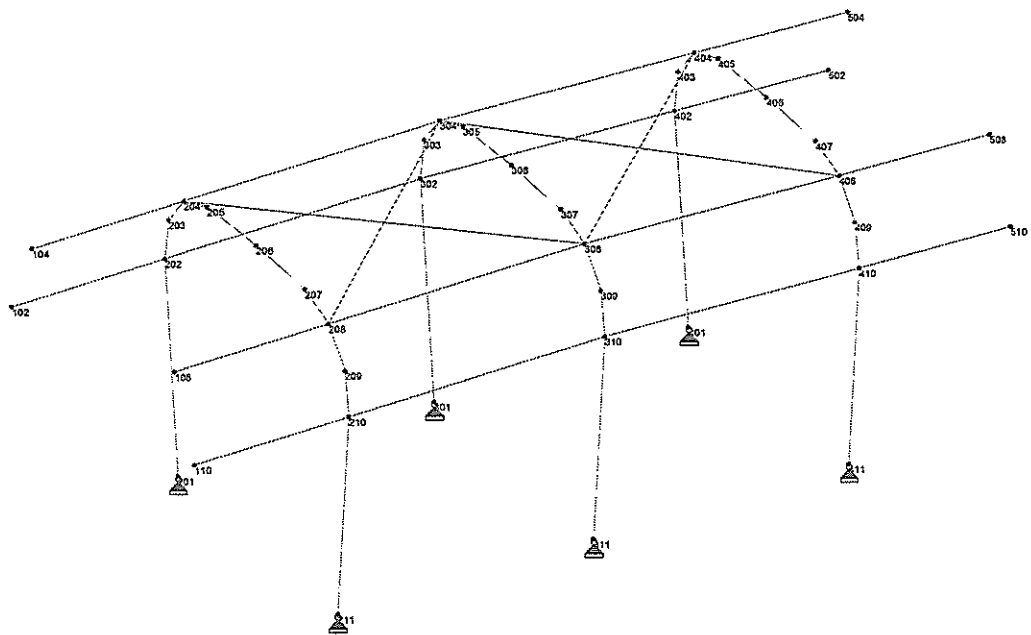
Load per column = $6 \times 12.6 / 6$
= 12.6 kN





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Part	Ref	
By	Date April 2007	Chd
Client	File Electrical Store.std	Date/Time 28-Apr-2007 15:18



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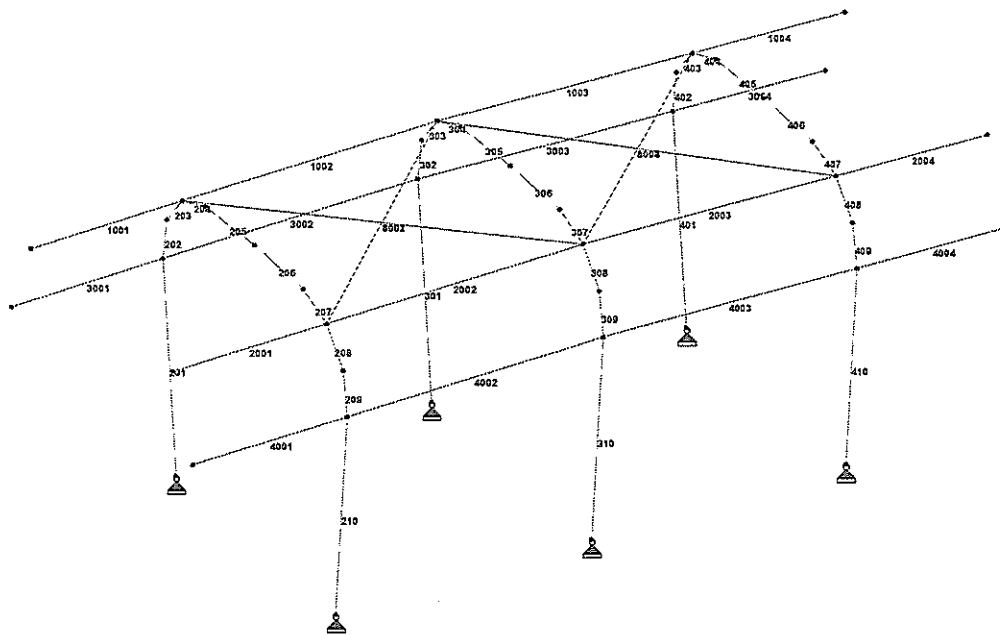
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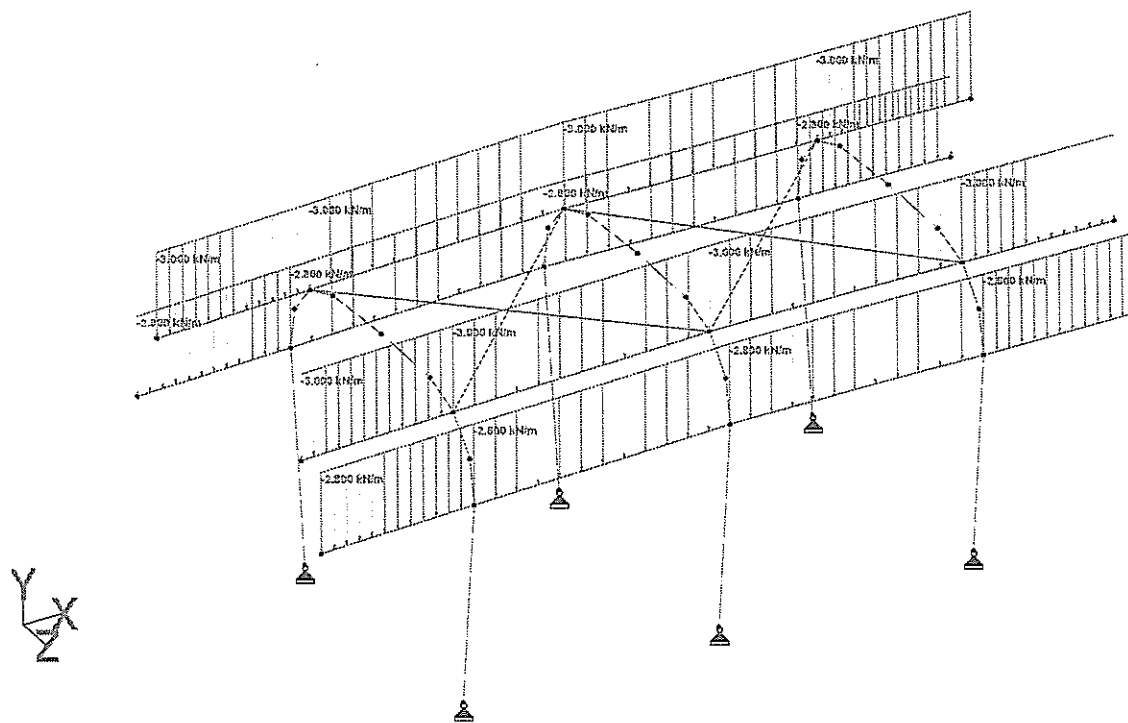
Date April 2007

Chd

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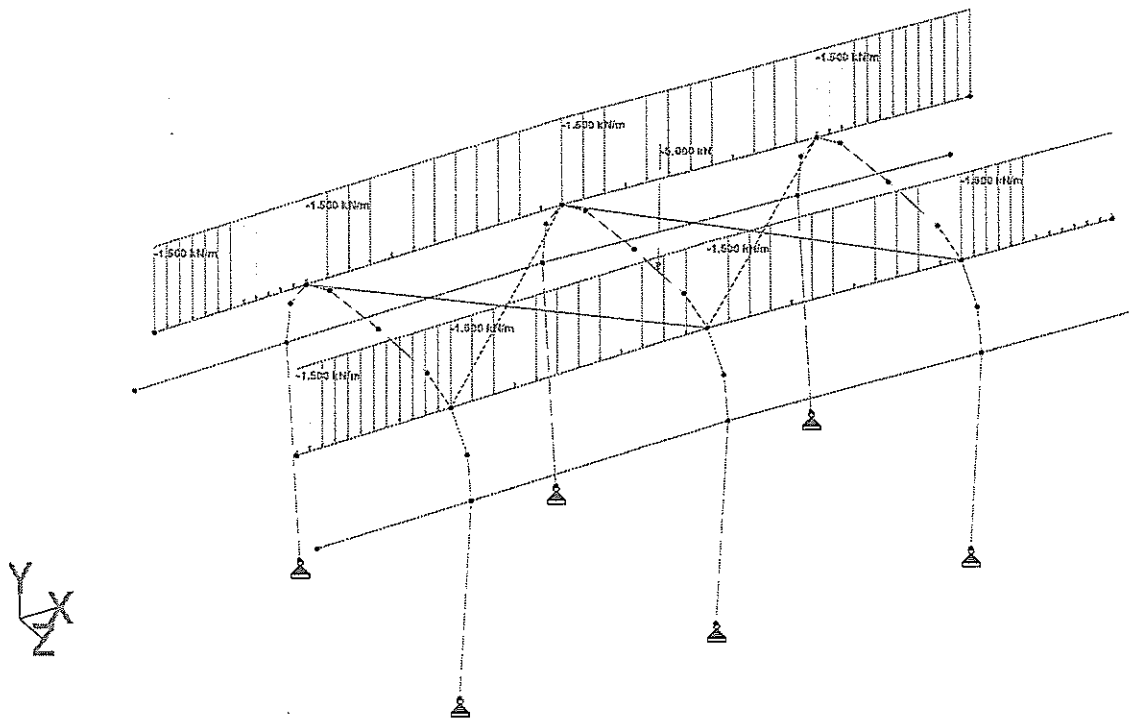


Load 2



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Part		
Job Title	Ref	
By	Date April 2007	Chd
Client	File Electrical Store.std	Date/Time 28-Apr-2007 15:18



Load 3



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Job No

Sheet No

1

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Part

Job Title

Ref

By

Date April 2007

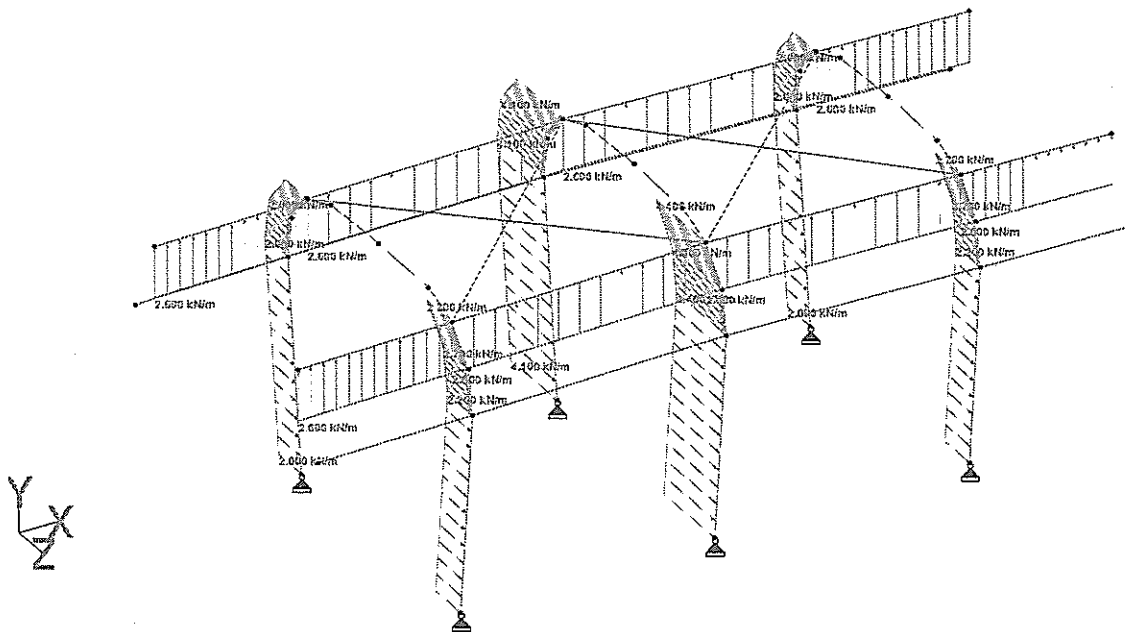
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Client

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Date/Time

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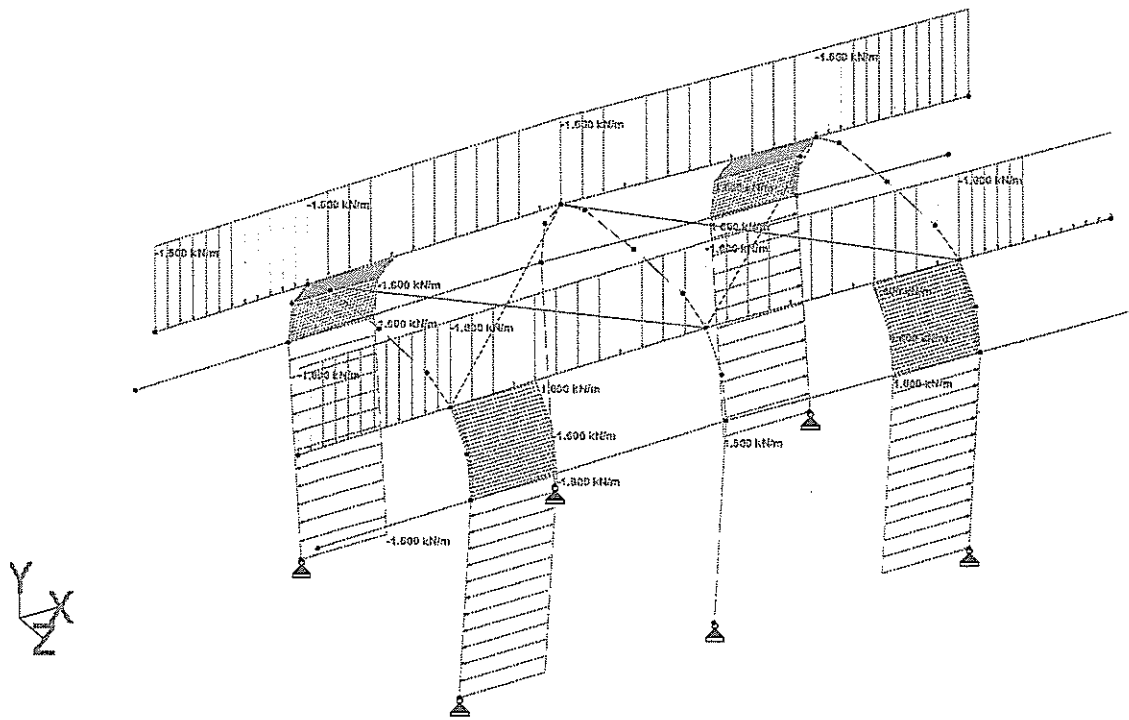


Load 12



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Job No	Sheet No 1	Rev
Part		
Ref		
By	Date April 2007	Chd
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Load 13



Job No	
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Sheet No

1

Rev

Part

Job Title

Ref

By _____

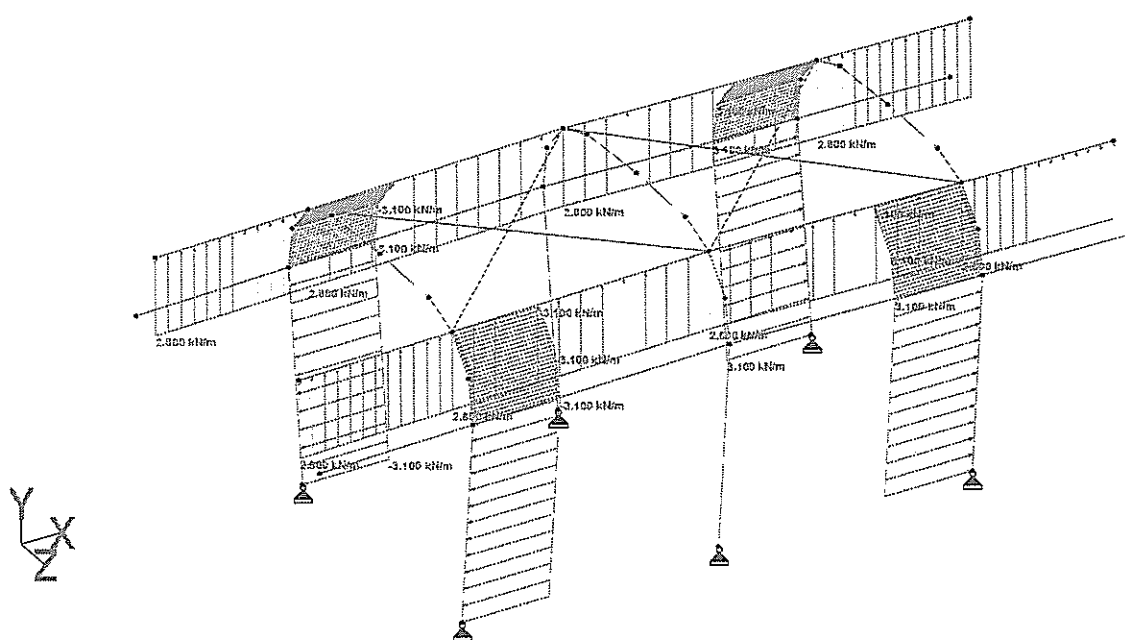
Date April 2007

Chd

Client	
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File Electrical Store.std

Date/Time 28-Apr-2007 15:18



Load 14



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Job No

Sheet No

1

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Part

Job Title

Ref

By

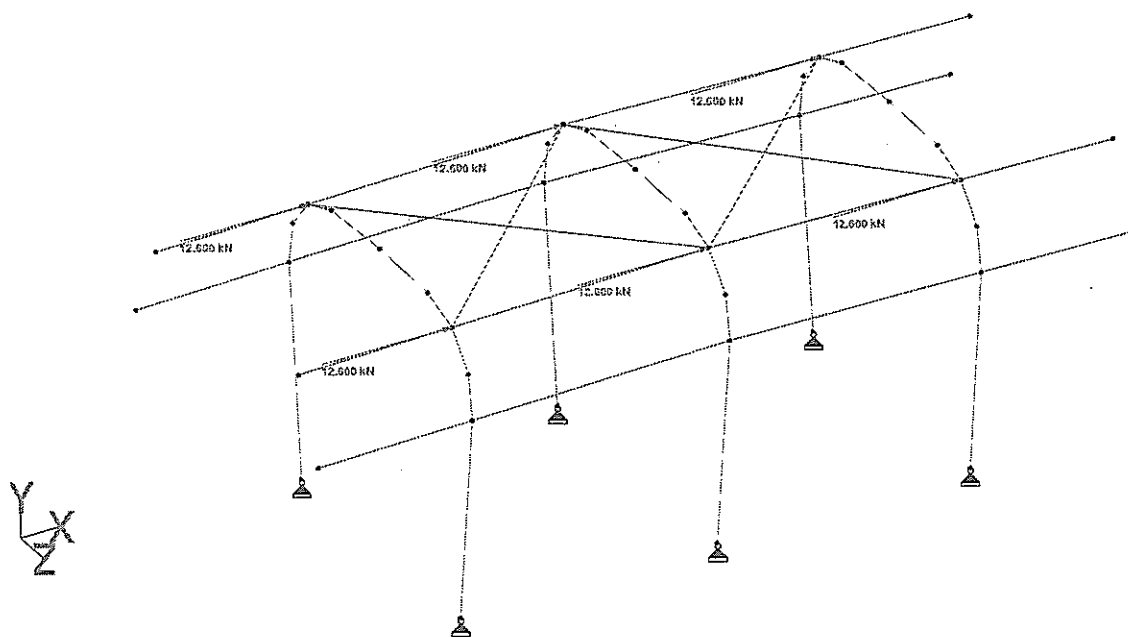
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Client

File Electrical Store.std

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Load 21



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Job No

Sheet No

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Part

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By

Date April 2007

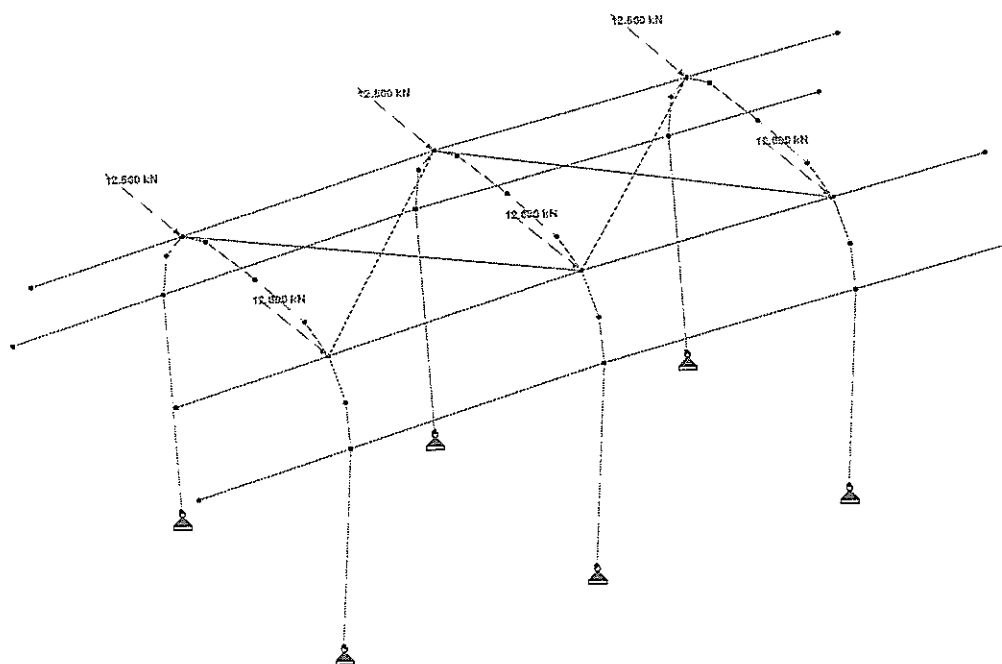
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Date/Time 28-Apr-2007 15:18

Job Title

Client



Load 22

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* Version 2006 81d 1002.US
* Proprietary Program of
* Research Engineers, Intl.
* Date= APR 28, 2007
* Time= 15:20:39
* USER ID: Meinhardt Infrastructure Pte Ltd

1. STAAD SPACE

INPUT FILE: Electrical Store.STD

2. START JOB INFORMATION

3. ENGINEER DATE APRIL 2007

4. END JOB INFORMATION

5. INPUT WIDTH 79

6. UNIT METER KN

7. JOINT COORDINATES

8. 102 -8.000 3.982 -3.102; 104 -8.000 5.464 -2.422
9. 110 -8.000 3.982 -3.102; 502 -8.000 3.982 -3.102
10. 308 8.000 3.464 -2.422; 108 -8.000 3.464 -2.422
11. 308 8.000 3.464 -2.422; 510 8.000 3.982 0.115
12. 308 8.000 3.464 -2.422; 502 -8.000 3.982 0.115
13. 308 8.000 3.464 -2.422; 502 -8.000 3.982 0.115
14. 308 8.000 3.464 -2.422; 502 -8.000 3.982 0.115
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18. 211 -5.000 0.000 2.685; 302 0.000 4.174 -3.124
20. 301 0.000 0.000 -2.685; 304 0.000 -2.440
21. 303 0.000 5.011 -2.986; 306 0.000 6.000 0.000
22. 305 0.000 5.943 -1.639; 308 0.000 5.662 2.440
23. 307 0.000 5.943 -1.639; 308 0.000 5.662 2.440
24. 309 0.000 5.011 -2.986; 310 0.000 4.174 3.124
25. 311 0.000 0.000 2.685; 402 5.000 4.054 -3.111
27. 401 0.000 0.000 -2.685; 404 5.000 3.328 -2.429
28. 403 0.000 5.889 -2.974; 406 5.000 5.875 0.000
29. 405 0.000 5.818 -1.631; 408 5.000 5.538 2.429
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31. 409 0.000 4.889 -2.974; 410 5.000 4.054 -3.111
32. 411 0.000 0.000 2.685
34. MEMBER INCIDENCES
35. 201 201 202 210; 301 301 302 310; 401 401 402 410
36. *

37. 1001 104 204; 1002 204 304; 1003 304 404; 1004 404 504
38. 2001 108 208; 2002 208 308; 2003 308 408; 2004 408 508
39. *
40. 3001 102 202; 3002 202 302; 3003 302 402; 3004 402 502
41. 4001 110 210; 4002 210 310; 4003 310 410; 4004 410 510
42. *
43. 8001 204 308; 8002 208 304; 8003 304 408; 8004 308 404
44. DEFINE MATERIAL START

STAAD SPACE

45. ISOTROPIC STEEL
46. E 2.05E+08
47. POISSON 0.3
48. DENSITY 76.8195
49. ALPHA 1.2E-005
50. DAMP 0.03
51. END DEFINE MATERIAL
53. MEMBER PROPERTY BRITISH
54. 201 TO 210 301 TO 310 401 TO 410 -
55. 1002 2002 2003 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
56. 1001 1004 2001 2004 TAPERED 0.3 0.006 0.3 0.15 0.008 0.15 0.008
57. 3001 TO 3004 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
58. 4001 TO 4004 TAPERED 0.3 0.010 0.3 0.25 0.015 0.25 0.015
59. 8001 TO 8004 TABLE ST UAP03X06
60. CONSTANT MEMBER 3001 TO 3004 4001 TO 4004
61. MATERIAL STEEL ALL
62. MEMBER TENSION
63. 8001 TO 8004
64. 8001 TO 8004
65. SUPPORTS
66. 201 211 301 311 401 411 PINNED
67. LOAD 1 SELFWEIGHT
68. SELFWEIGHT Y -1
69. LOAD 2 SDL
70. MEMBER LOAD

71. 1001 TO 1004 2001 TO 2004 UNI GY -3.0
72. 3001 TO 3004 4001 TO 4004 UNI GY -2.8
73. LOAD 3 LL
74. MEMBER LOAD
75. 1001 TO 1004 2001 TO 2004 UNI GY -1.5
76. 306 CON Y -5
77. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)
78. MEMBER LOAD
79. 1001 TO 1004 2001 TO 2004 UNI GY 1.9
80. 201 TO 203 401 TO 403 UNI GZ 3.6
81. 301 TO 303 401 TO 403 UNI GZ 0.6
82. 308 TO 310 UNI GZ 1.2
83. 308 TO 310 UNI GZ 1.2
84. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)
85. MEMBER LOAD
86. 1001 TO 1004 2001 TO 2004 UNI GY 2.6
87. 201 TO 203 401 TO 403 UNI GZ 2.0
88. 301 TO 303 401 TO 403 UNI GZ 4.1
89. 308 TO 310 UNI GZ 4.4
90. 308 TO 310 UNI GZ 4.4
91. 308 TO 310 UNI GZ 4.4
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94. 1001 TO 1004 2001 TO 2004 UNI GY -1.6
95. 201 TO 203 401 TO 403 UNI GZ 1.6
96. 301 TO 303 401 TO 403 UNI GZ 1.6
97. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)
98. MEMBER LOAD
99. 1001 TO 1004 2001 TO 2004 UNI GY 2.8
100. 201 TO 203 401 TO 403 UNI GZ 3.1
101. 301 TO 303 401 TO 403 UNI GZ 3.1
102. 308 TO 310 UNI GZ 3.1
103. 308 TO 310 UNI GZ 3.1
104. LOAD 21 EQ1 (EARTHQUAKE LOAD)
105. STAAD SPACE
107. JOINT LOAD
108. 204 304 308 404 408 FX 12.6
109. LOAD 22 EQ1 (EARTHQUAKE LOAD)
110. JOINT LOAD
111. 204 308 304 308 404 408 FZ 12.6
112. LOAD COMB 1001 1.4DL + 1.6LL
113. 1.4 2 1.4 3 1.6
114. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1
115. 1.2 2 1.1 1.1 1.2 1.1 1.4
116. 1.4 2 1.1 1.1 1.2 1.1 1.4
117. 1.4 2 1.1 1.1 1.2 1.1 1.4
118. 1.4 2 1.1 1.1 1.2 1.1 1.4
119. 1.4 2 1.1 1.1 1.2 1.1 1.4
120. LOAD COMB 1211 1.0DL + 1.4WL1
121. 1.0 2 1.0 1.1 1.4
122. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2
123. 1.2 2 1.2 3 1.2 1.2 1.2
124. LOAD COMB 1112 1.4DL + 1.4WL2
125. 1.4 2 1.4 1.2 1.4
126. LOAD COMB 1212 1.0DL + 1.4WL2
127. 1.0 2 1.0 1.2 1.4
128. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3
129. 1.2 2 1.2 3 1.2 1.2 1.2
130. LOAD COMB 1113 1.4DL + 1.4WL3
131. 1.4 2 1.1 1.1 1.2 1.1 1.4
132. 1.4 2 1.1 1.1 1.2 1.1 1.4
133. 1.4 2 1.1 1.1 1.2 1.1 1.4
134. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4
135. 1.2 2 1.2 3 1.2 1.2 1.2
136. LOAD COMB 1114 1.4DL + 1.4WL4
137. 1.4 2 1.4 1.4 1.4
138. LOAD COMB 1214 1.0DL + 1.4WL4
139. 1.0 2 1.0 1.4 1.4
140. 1.0 2 1.0 1.4 1.4
141. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ1
142. 1.2 2 1.2 3 0.5 2.1 1.0
143. LOAD COMB 1022 1.2DL + 0.5LL + 1.0EQ2
144. 1.2 2 1.2 3 0.5 2.1 1.0
145. LOAD COMB 1121 1.2DL + 0.5LL + 1.0EQ1
146. 1.2 2 1.2 3 0.5 2.1 1.0
147. 1.2 2 1.2 3 0.5 2.1 1.0
148. LOAD COMB 1221 0.9DL + 1.0EQ1
149. 0.9 2 0.9 2.1 1.0
150. 0.9 2 0.9 2.1 1.0
151. LOAD COMB 1022 0.9DL + 1.0EQ2
152. 0.9 2 0.9 2.1 1.0
153. LOAD COMB 1321 0.9DL + 1.0EQ1
154. 0.9 2 0.9 2.1 1.0
155. LOAD COMB 1322 0.9DL + 1.0EQ2
156. 0.9 2 0.9 2.1 1.0
157. *SLS
158. LOAD COMB 2001 1.0DL + 1.0LL
159. 1.0 2 1.0 2.1 1.0
160. 1.0 2 1.0 2.1 1.0
161. 1.0 2 1.0 2.1 1.0
162. LOAD COMB 2011 0L + LL + WL1

Electrical Store.ANL

Electrical Store.ANL

163. 1 1.0 2 1.0 3 1.0 11 1.0
164. LOAD COMB 2012 DL + LL + WL2
165. 1 1.0 2 1.0 3 1.0 12 1.0
STAAD SPACE

--- PAGE NO. 4

166. LOAD COMB 2013 DL + LL + WL3
167. 1 1.0 2 1.0 3 1.0 13 1.0
168. LOAD COMB 2014 DL + LL + WL4
169. 1 1.0 2 1.0 3 1.0 14 1.0
171. PERFORM ANALYSIS

PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER-ELEMENTS/SUPPORTS = 41/ 50/
ORIGINAL/FINAL BAND-WIDTH= 32/ 11/
TOTAL DEGREES OF FREEDOM = 228
TOTAL PRIMARY LOAD CASES = 9, 17 DOUBLE KILO-WORDS
SIZE OF STIFFNESS MATRIX =
REQD/AVAIL. DISK SPACE = 12.3/ 416818.9 MB

**NOTE-Tension/Compression converged after 1 iterations, Case= 1

**NOTE-Tension/Compression converged after 1 iterations, Case= 2

**NOTE-Tension/Compression converged after 1 iterations, Case= 3

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 11

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 12

**NOTE-Tension/Compression converged after 1 iterations, Case= 13

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 14

**START ITERATION NO. 2
STAAD SPACE

--- PAGE NO. 5

**NOTE-Tension/Compression converged after 2 iterations, Case= 21

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 22

172. LOAD LIST 1001 1011 TO 1014 1021 1022 1111 TO 1114 1121 1122 -
173. 1211 TO 1214 1221 1222 1321 1322
174. PRINT SUPPORT REACTION ALL
SUPPORT REACTION ALL
STAAD SPACE

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JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MON-X	MON-Y	MON-Z
201	1001	-0.23	77.43	-2.94	0.00	0.00	0.00
	1011	-0.24	26.34	-25.27	0.00	0.00	0.00
	1111	-0.21	29.04	-29.44	0.00	0.00	0.00
	1211	-0.21	23.17	-28.61	0.00	0.00	0.00
	1012	-0.21	23.25	-20.16	0.00	0.00	0.00
	1112	-0.25	14.10	-23.47	0.00	0.00	0.00
	1212	-0.18	-3.77	-22.64	0.00	0.00	0.00
	1013	2.81	75.56	-2.52	0.00	0.00	0.00
	1113	3.27	75.13	-2.90	0.00	0.00	0.00
	1213	3.34	57.26	-2.07	0.00	0.00	0.00
	1014	5.60	42.45	-2.23	0.00	0.00	0.00
	1114	6.53	36.51	-2.56	0.00	0.00	0.00
	1214	6.60	18.64	-1.73	0.00	0.00	0.00
	1021	-12.79	37.17	-0.75	0.00	0.00	0.00
	1022	-0.21	31.52	-13.24	0.00	0.00	0.00
	1121	12.38	89.01	-10.51	0.00	0.00	0.00
	1221	-12.74	19.12	-0.11	0.00	0.00	0.00
	1023	-0.16	13.46	-14.66	0.00	0.00	0.00
	1321	12.43	61.30	-3.61	0.00	0.00	0.00
	1322	-0.15	66.96	10.94	0.00	0.00	0.00
211	1001	-0.23	77.24	3.03	0.00	0.00	0.00
	1011	-0.21	72.68	-10.45	0.00	0.00	0.00
	1111	-0.25	71.94	-12.31	0.00	0.00	0.00
	1211	-0.18	54.07	-13.13	0.00	0.00	0.00
	1012	-0.22	68.01	-15.67	0.00	0.00	0.00
	1112	-0.26	66.49	-18.40	0.00	0.00	0.00
	1212	-0.19	48.62	-19.22	0.00	0.00	0.00
	1013	2.81	75.41	2.58	0.00	0.00	0.00
	1113	3.27	75.13	2.90	0.00	0.00	0.00
	1213	3.34	57.26	2.07	0.00	0.00	0.00
	1014	5.60	42.45	2.23	0.00	0.00	0.00
	1114	6.53	36.51	2.56	0.00	0.00	0.00
	1214	6.60	18.64	1.73	0.00	0.00	0.00
	1021	-12.79	37.17	0.75	0.00	0.00	0.00
	1022	-0.21	31.52	13.24	0.00	0.00	0.00
	1121	12.38	89.01	10.51	0.00	0.00	0.00
	1221	-12.74	19.12	0.11	0.00	0.00	0.00
	1023	-0.16	13.46	14.61	0.00	0.00	0.00
301	1001	0.00	53.79	-1.67	0.00	0.00	0.00
	1011	0.00	18.05	-33.76	0.00	0.00	0.00
	1111	0.00	10.79	-39.51	0.00	0.00	0.00
	1211	0.00	1.42	-34.81	0.00	0.00	0.00
	1012	0.00	14.41	-24.46	0.00	0.00	0.00
	1112	0.00	6.54	-28.66	0.00	0.00	0.00
STAAD SPACE		0.00	0.00				

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--- PAGE NO.

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MON-X	MON-Y	MON-Z
1	1212	0.00	5.48	-28.14	0.00	0.00	0.00
	1312	0.00	53.01	-1.71	0.00	0.00	0.00
	1013	0.00	57.25	-2.12	0.00	0.00	0.00
	1113	0.00	40.73	-1.60	0.00	0.00	0.00
	1014	0.00	35.71	-1.98	0.00	0.00	0.00
	1114	0.00	31.40	-2.43	0.00	0.00	0.00
	1214	0.00	19.38	-1.91	0.00	0.00	0.00
	1021	-12.64	39.72	-1.53	0.00	0.00	0.00
	1022	0.00	14.66	-13.80	0.00	0.00	0.00
	1121	12.64	39.71	-1.49	0.00	0.00	0.00
	1221	0.00	64.76	10.78	0.00	0.00	0.00
	1023	-12.64	27.04	-1.18	0.00	0.00	0.00
	1222	0.00	2.38	-1.13	0.00	0.00	0.00
	1321	12.64	57.08	11.13	0.00	0.00	0.00
	1322	0.00	51.99	1.78	0.00	0.00	0.00
311	1001	0.00	57.09	-12.18	0.00	0.00	0.00
	1011	0.00	54.41	-14.18	0.00	0.00	0.00
	1111	0.00	42.39	-14.70	0.00	0.00	0.00
	1211	0.00	53.36	-21.97	0.00	0.00	0.00
	1012	0.00	50.06	-25.61	0.00	0.00	0.00
1112		0.00					

Page 4

201	0.00	MAX	19.13	1122	0.00	1022	12.78	1321	0.00	1321
	0.82	MIN	-28.23	1221	0.90	1111	1.43	1311	0.00	1021
		MAX	24.19	1211	21.56	1121	1.43	1311	10.13	1321
	1.63	MIN	19.07	1121	-11.56	1121	1.78	1021	-10.42	1021
		MAX	19.07	1121	39.40	1121	1.43	1311	20.26	1321
	2.45	MIN	-20.11	1211	-31.14	1122	11.79	1021	20.85	1021
		MAX	19.04	1122	54.14	1211	12.43	1321	30.40	1321
	3.26	MIN	-16.18	1212	-46.68	1122	12.79	1021	-31.27	1021
		MAX	19.00	1021	65.57	1211	12.43	1321	40.53	1321
	4.08	MIN	-13.94	1212	-62.18	1122	11.79	1021	-41.70	1021
		MAX	18.97	1122	73.69	1212	12.43	1321	50.66	1321
		MIN	-13.29	1222	-77.46	1122	11.79	1021	-52.12	1021

	MAX/MIN FORCE VALUES FOR MEMB				201. AMONGST ALL SECT LOCATIONS			
	FZ/ FY	DIST LD	DIST LD	NY	DIST LD	DIST LD	FX	DIST LD
MAX.	19.13	0.00	1122	73.69	4.08	1211	83.47	C 0.00 1122
	12.43	0.00	1321	50.66	4.08	1321		
MIN.	-28.23	0.00	1211	-52.12	4.08	1222	3.85	T 4.08 1212
	-12.79	4.08	1021	-57.12	4.08	1021		

202	0.00	MAX	15.87	1322	77.64	1122	11.12	1021	5.89	1321
	0.17	MIN	15.86	1322	77.66	1122	11.12	1021	-9.19	1321
	0.34	MAX	15.86	1322	77.99	1122	11.12	1021	4.66	1321
		MIN	15.86	1322	77.99	1122	11.12	1021	-7.31	1021
	0.51	MAX	15.85	1022	-74.05	1122	-7.30	1321	3.42	1321
		MIN	15.85	1022	-74.05	1122	-7.30	1321	-5.43	1021
	0.68	MAX	15.84	1022	-74.26	1212	-7.30	1321	2.19	1321
		MIN	15.84	1022	-74.26	1212	-7.30	1321	-3.52	1021
	0.85	MAX	15.83	1022	-74.33	1212	-7.30	1321	0.62	1321
		MIN	15.83	1022	-74.33	1212	-7.30	1321	-1.62	1021
	STAAD SPACE	MAX	15.82	1022	-79.06	1122	-1.76	1321	0.22	1321
		MIN	15.82	1022	-79.06	1122	-1.76	1321	-0.22	1321
	STAAD SPACE	MAX	-4.43	1322	-74.66	1212	-7.30	1321	-0.29	1321
		MIN	-4.43	1322	-74.66	1212	-7.30	1321	0.29	1321

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MAX/MIN FORCE VALUES FOR MEMB						202, AMONGST ALL SECT LOCATIONS							
	FY/ FZ	DIST LD	DIST LD	MZ/ MY	DIST LD	DIST LD	FX	DIST LD					
MAX.	15.87	0.00	10221	79.42	0.85	1122							
	11.12	0.00	1021	5.69	0.00	1321							
MIN.	-7.43	0.85	1322	-74.66	0.85	1212							
	-7.30	0.85	1321	-9.19	0.00	1021							
							58.64	C	0.00	1122			
							28.15	T	0.00	1212			

2003	0.00	MAX	32.81	1013	79.42	1122	11.32	1021	0.20	1321
		MIN	-12.19	1211	-74.86	1112	11.32	1021	-0.57	1321
	0.17	MAX	32.78	1011	-74.98	1112	11.32	1021	2.09	1021
		MIN	-22.98	1111	62.94	1123	11.30	1031	-1.50	1321
	0.34	MAX	32.55	1211	-70.79	1213	11.32	1021	3.97	1021
		MIN	-22.57	1111	70.79	1213	11.32	1021	-2.74	1321
	0.51	MAX	32.68	1012	65.98	1323	11.32	1031	5.86	1021
		MIN	-14.26	1211	-69.49	1323	11.32	1031	-3.98	1321
	0.68	MAX	32.64	1013	62.54	1322	11.12	1021	7.74	1021
		MIN	-14.94	1211	-71.87	1022	11.32	1031	-5.21	1321
	0.85	MAX	32.60	1013	59.11	1322	11.12	1021	9.63	1021
		MIN	-15.63	1211	-74.24	1022	11.32	1031	-6.45	1321

MAX/MIN FORCE VALUES FOR MEMB				203, AMONGST ALL SECT LOCATIONS					
	F _y /F _z	DIST	LD	F _y /F _z	DIST	LD	FX	DIST	LD
MAX.	32.81	0.00	1013	79.42	0.00	1122			
	11.12	0.00	1021	9.63	0.85	1021			
MIN.	-15.63	0.85	1311	-74.66	0.00	1312	51.23	C	0.00 1322
	-17.30	0.85	1321	-6.45	0.85	1331	25.98	Y	0.00 1312

2004	0.00	MAX	24.75	1322	59.08	1322	0.00	1211	0.26	1021
		MIN	-28.67	1111	-74.27	1022	0.00	1022	-0.26	1321
	0.17	MAX	24.71	1322	54.90	1322	0.00	1211	0.26	1021
		MIN	-28.75	1111	-69.71	1022	0.00	1022	-0.13	1321

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-2-30

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-0-

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STAAD

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SUPPORT REACTIONS -UNIT KN      METE      STRUCTURE TYPE = SPACE
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JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM-Z
1214		-6.60	18.64	-1.73	0.00	0.00	0.00
1021		-12.38	79.35	-4.27	0.00	0.00	0.00
1022		0.21	31.52	-15.29	0.00	0.00	0.00
1121		12.79	37.18	0.72	0.00	0.00	0.00
1122		0.20	85.01	10.31	0.00	0.00	0.00
1221		-12.43	61.30	-3.64	0.00	0.00	0.00
1222		0.16	13.46	-14.66	0.00	0.00	0.00
1321		12.74	19.12	-0.08	0.00	0.00	0.00
1322		0.15	66.96	10.94	0.00	0.00	0.00
411	1001	0.23	77.24	3.03	0.00	0.00	0.00
1011		0.21	72.68	-10.45	0.00	0.00	0.00
1111		0.25	71.94	-12.31	0.00	0.00	0.00
1211		0.18	54.07	-13.13	0.00	0.00	0.00
1012		0.22	68.01	-15.67	0.00	0.00	0.00
1112		0.26	66.49	-18.40	0.00	0.00	0.00
1212		0.19	48.62	-19.22	0.00	0.00	0.00
1013		-2.81	75.41	2.58	0.00	0.00	0.00
1113		-3.27	75.13	2.90	0.00	0.00	0.00
1213		-3.34	57.26	2.07	0.00	0.00	0.00
1014		-5.60	42.51	2.29	0.00	0.00	0.00
1114		-6.83	36.31	2.36	0.00	0.00	0.00
1214		-7.98	28.94	1.33	0.00	0.00	0.00
1023		-6.20	81.63	-1.33	0.00	0.00	0.00
1123		17.29	37.17	0.72	0.00	0.00	0.00
1223		16.71	31.47	15.27	0.00	0.00	0.00
1323		-2.43	61.30	3.64	0.00	0.00	0.00
1222		0.15	66.95	-10.89	0.00	0.00	0.00
1321		12.74	19.12	0.08	0.00	0.00	0.00
1322		0.16	13.48	14.61	0.00	0.00	0.00

***** END OF LATEST ANALYSTS RESULT *****

175. PRINT FORCE ENVELOPE NSECTION 5 ALL
FORCE ENVELOPE NSECTION 5

5

PAGE NO.

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300
VELOPE
SN 3007

ORCE
STARS

MEMBER FORCE ENVELOPE

ALL UNITS ARE KN METE

MEMB	DISTANCE	FY	LD	NZ	LD	FZ	LD	NY	LD
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Electrical Store ANI
0.00 1211 0.26 1021
58.95 1211 0.00 1022
64.07 1211 0.00 1022
64.07 1211 0.00 1022
74.22 1122 0.00 1022

0.68 MAX 26.49 1122
MIN 26.49 1122
0.85 MIN -24.17 1122

MAX/MIN FORCE VALUES FOR MEMB
FY/ DIST LD MZ/ DIST LD MY/ DIST LD FX DIST LD

207, AMONGST ALL SECT LOCATIONS
MAX. 26.73 0.00 1122 62.07 0.85 1211 24.67 C 0.85 1111
MIN. -24.17 0.85 1222 -74.22 0.85 1122 5.49 T 0.00 1322

208 0.00 MAX 0.35 1214 62.08 1211 7.30 1321 9.62 1021
MIN -32.63 1013 -74.19 1122 -11.11 1021 -6.45 1321
0.17 MAX 0.31 1214 64.27 1211 7.30 1321 7.73 1021
MIN -32.68 1013 -71.82 1122 -11.11 1021 -5.21 1321
0.34 MAX 0.28 1214 66.45 1211 7.30 1321 5.85 1021
MIN -32.72 1013 -69.45 1122 -11.11 1021 -3.98 1321
0.51 MAX 0.25 1214 69.54 1022 7.30 1321 3.97 1021
MIN -32.76 1013 -67.49 1322 -11.11 1021 -2.74 1321
0.68 MAX 0.21 1214 74.27 1022 7.30 1321 2.08 1021
MIN -32.80 1013 -66.29 1322 -11.11 1021 -1.50 1321
0.85 MAX 0.18 1214 79.01 1022 7.30 1321 0.27 1121
MIN -32.84 1013 -65.17 1322 -11.11 1021 -0.27 1121

STAAD SPACE

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MAX/MIN FORCE VALUES FOR MEMB
FY/ DIST LD MZ/ DIST LD MY/ DIST LD FX DIST LD

208, AMONGST ALL SECT LOCATIONS
MAX. 0.35 0.00 1214 79.01 0.85 1022 51.11 C 0.85 1022
MIN. -32.84 0.85 1013 -74.19 0.00 1122 15.47 T 0.00 1322

209 0.00 MAX 4.41 1222 79.01 1022 7.30 1321 0.22 1221
MIN -15.86 1122 -65.17 1322 -11.11 1021 -0.29 1121
0.17 MAX 4.40 1222 78.66 1022 7.30 1321 0.95 1321
MIN -15.87 1122 -62.88 1322 -11.11 1021 -1.66 1021
0.34 MAX 4.52 1211 78.32 1022 7.30 1321 2.19 1321
MIN -15.88 1122 -60.60 1322 -11.11 1021 -3.54 1021
0.51 MAX 4.65 1211 77.98 1022 7.30 1321 3.42 1321
MIN -15.89 1122 -58.31 1322 -11.11 1021 -5.42 1021
0.68 MAX 4.78 1211 77.64 1022 7.30 1321 4.66 1321
MIN -15.90 1122 -56.02 1322 -11.11 1021 -7.30 1321
0.85 MAX 4.91 1211 77.30 1022 7.30 1321 5.18 1021
MIN -15.91 1122 -53.73 1322 -11.11 1021 -9.18 1021

MAX/MIN FORCE VALUES FOR MEMB
FY/ DIST LD MZ/ DIST LD MY/ DIST LD FX DIST LD

209, AMONGST ALL SECT LOCATIONS
MAX. 4.91 0.85 1211 79.01 0.00 1022 58.55 C 0.85 1022
MIN. -15.91 0.85 1122 -65.17 0.00 1322 9.77 T 0.00 1322

210 0.00 MAX 13.24 1322 53.72 1322 12.79 1021 50.66 1321
MIN -13.24 1322 -72.72 1022 -12.43 1321 -52.12 1021
0.82 MAX 13.24 1322 42.84 1322 12.79 1021 40.53 1321
MIN -18.95 1022 -65.79 1112 -12.43 1321 -41.70 1021
1.63 MAX 13.19 1322 32.17 1322 12.79 1021 30.40 1321
MIN -18.95 1022 -52.45 1112 -12.43 1321 -31.27 1021
2.45 MAX 13.17 1322 21.43 1322 12.79 1021 20.26 1321
MIN -20.18 1112 -37.03 1112 -12.43 1321 -20.85 1021
3.26 MAX 13.14 1322 10.71 1322 12.79 1021 10.13 1321
MIN -22.71 1112 -19.55 1112 -12.43 1321 -0.00 1321
4.08 MAX 13.12 1322 0.00 1013 -12.43 1321 0.00 1122
MIN -25.25 1112 0.00 1212 -12.43 1321 0.00 1122

STAAD SPACE

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Electrical Store ANI
0.00 1211 0.26 1021
50.72 1322 0.00 1022
-65.14 1022 0.00 1022
-65.15 1322 0.00 1321 0.16 1321
42.39 1322 0.00 1211 0.26 1021
-55.96 1022 0.00 1022 -0.13 1321
38.24 1322 0.00 1211 0.26 1021
-51.36 1022 0.00 1022 -0.13 1321

0.34 MAX 24.66 1322
MIN -28.82 1111
0.51 MAX 24.82 1322
MIN -28.82 1111
0.68 MAX 24.57 1322
MIN -28.96 1111
0.85 MAX 24.53 1322
MIN -29.03 1111

STAAD SPACE

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MAX/MIN FORCE VALUES FOR MEMB
FY/ DIST LD MZ/ DIST LD MY/ DIST LD FX DIST LD

204, AMONGST ALL SECT LOCATIONS
MAX. 24.75 0.00 1322 59.08 0.00 1322 15.05 C 0.00 1122
MIN. -29.03 0.85 1111 -74.27 0.00 1022 3.82 T 0.85 1222
0.00 0.85 1022 -0.13 0.85 1321

205 0.00 MAX 27.14 1122 38.24 1322 0.00 1211 0.27 1021
MIN -26.42 1222 -51.36 1022 0.00 1022 -0.14 1321
0.33 MAX 27.02 1322 29.43 1322 0.00 1211 0.27 1021
MIN -26.52 1222 -42.75 1022 0.00 1022 -0.14 1321
0.65 MAX 26.90 1322 20.64 1322 0.00 1211 0.27 1021
MIN -26.91 1222 -34.09 1022 0.00 1022 -0.14 1321
0.98 MAX 26.76 1322 -11.89 1322 0.00 1211 0.27 1021
MIN -26.76 1322 -23.19 1322 0.00 1211 0.27 1021
1.31 MAX 26.68 1322 -16.66 1022 0.00 1211 0.27 1021
MIN -26.82 1022 -16.66 1022 0.00 1211 0.27 1021
1.63 MAX 26.59 1322 3.97 1211 0.00 1211 -0.14 1321
MIN -26.95 1022 -11.85 1013 0.00 1022 -0.14 1321

MAX/MIN FORCE VALUES FOR MEMB
FY/ DIST LD MZ/ DIST LD MY/ DIST LD FX DIST LD

205, AMONGST ALL SECT LOCATIONS
MAX. 27.14 0.00 1122 38.24 0.00 1322 17.02 C 0.00 1011
MIN. -26.95 1.63 1022 -51.36 0.00 1022 1.63 C 1.63 1321
0.00 1.63 1022 -0.14 1.63 1321

206 0.00 MAX 26.96 1122 3.97 1211 0.00 1211 0.27 1021
MIN -26.53 1222 -11.85 1013 0.00 1022 -0.14 1321
0.33 MAX 26.84 1122 0.27 1211 0.00 1211 0.27 1021
MIN -26.62 1222 -17.03 1122 0.00 1022 -0.14 1321
0.65 MAX 26.72 1122 20.40 1211 0.00 1211 0.27 1021
MIN -26.71 1222 -25.77 1122 0.00 1022 -0.14 1321
0.98 MAX 26.59 1122 28.66 1211 0.00 1211 0.27 1021
MIN -26.80 1222 -34.47 1122 0.00 1022 -0.14 1321
1.31 MAX 26.47 1122 36.96 1211 0.00 1211 0.27 1021
MIN -26.90 1222 -43.13 1122 0.00 1022 -0.14 1321
1.63 MAX 26.36 1322 45.29 1211 0.00 1211 0.27 1021
MIN -27.00 1022 -51.74 1122 0.00 1022 -0.14 1321

STAAD SPACE

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MAX/MIN FORCE VALUES FOR MEMB
FY/ DIST LD MZ/ DIST LD MY/ DIST LD FX DIST LD

206, AMONGST ALL SECT LOCATIONS
MAX. 26.96 0.00 1122 45.29 1.63 1211 18.55 C 1.63 1011
MIN. -27.00 1.63 1022 -51.74 1.63 1122 1.63 C 0.00 1321
0.00 1.63 1022 -0.14 1.63 1321

207 0.00 MAX 26.73 1122 45.29 1211 0.00 1211 0.26 1021
MIN -26.94 1222 -51.74 1122 0.00 1022 -0.13 1321
0.17 MAX 26.67 1122 48.63 1211 0.00 1211 0.26 1021
MIN -23.99 1222 -56.26 1122 0.00 1022 -0.13 1321
0.34 MAX 26.61 1122 51.98 1211 0.00 1211 0.26 1021
MIN -24.03 1222 -60.77 1122 0.00 1022 -0.13 1321
0.51 MAX 26.55 1122 55.33 1211 0.00 1211 0.26 1021
MIN -24.08 1222 -65.26 1122 0.00 1022 -0.13 1321

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MAX.		23.57	0.00	1013	73.38	0.00	1122	40.52	C	0.00	1122
MIN.		-23.51	0.85	1211	-70.69	0.00	1211	17.69	T	0.85	1222
		-20.04	0.85	1321	-16.95	0.85	1321				
304	0.00	MAX	22.04	1322	56.07	1322	0.00	1321		0.26	1321
	MIN	22.00	1311	-68.82	1022	0.00	1021	0.00	1021	-0.26	1321
	0.17	MAX	22.00	1322	57.93	1322	0.00	1321		0.26	1321
	MIN	30.40	1111	-64.21	1022	0.00	1021	0.00	1021	-0.26	1321
	0.34	MAX	21.95	1322	48.60	1322	0.00	1321		0.26	1321
	MIN	30.47	1111	-59.58	1022	0.00	1021	0.00	1021	-0.26	1321
	0.51	MAX	21.91	1322	44.88	1322	0.00	1321		0.26	1321
	MIN	30.54	1111	-54.94	1022	0.00	1021	0.00	1021	-0.26	1321
	0.68	MAX	21.86	1322	41.17	1322	0.00	1321		0.26	1321
	MIN	30.61	1111	-50.29	1022	0.00	1021	0.00	1021	-0.26	1321
	0.85	MAX	21.81	1322	37.46	1322	0.00	1321		0.26	1321
	MIN	30.68	1111	-45.63	1022	0.00	1021	0.00	1021	-0.26	1321
STAAD SPACE											
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304, AMONGST ALL SECT LOCATIONS											
MAX/MIN FORCE VALUES FOR MEMB			NZ/			DIST			LD		
FY/	FZ/		DIST	LD		DIST	LD		DIST	LD	
MAX.	22.04	0.00	1322		56.07	0.00	1322		26.96	C	0.00
	0.00	0.00	1321		0.26	0.00	1022				0.00
MIN.	-30.68	0.85	1111		-68.82	0.00	1022		1.45	C	0.85
	0.00	0.85	1021		-0.26	0.85	1321				0.85

305, AMONGST ALL SECT LOCATIONS											
MAX/MIN FORCE VALUES FOR MEMB			NZ/			DIST			LD		
FY/	FZ/		DIST	LD		DIST	LD		DIST	LD	
MAX.	0.00	MAX	26.06	1122	37.46	1322	0.00	1221		0.28	1221
	MIN	24.89	1222	-45.63	1022	0.00	1121	0.00	1121	-0.28	1321
	0.33	MAX	25.94	1122	29.20	1322	0.00	1221		0.28	1221
	MIN	24.98	1222	-37.72	1022	0.00	1121	0.00	1121	-0.28	1321
	0.66	MAX	25.81	1122	20.97	1322	0.00	1221		0.28	1221
	MIN	25.07	1222	-29.77	1022	0.00	1121	0.00	1121	-0.28	1321
	0.98	MAX	25.69	1122	12.77	1322	0.00	1221		0.28	1221
	MIN	25.16	1222	-21.78	1022	0.00	1121	0.00	1121	-0.28	1321
	1.31	MAX	25.57	1122	4.90	1322	0.00	1221		0.28	1221
	MIN	25.26	1222	-8.68	1022	0.00	1121	0.00	1121	-0.28	1321
	1.64	MAX	25.44	1122	-10.48	1001	0.00	1221		0.28	1221
	MIN	25.33	1222			0.00	1121	0.00	1121	-0.28	1321

305, AMONGST ALL SECT LOCATIONS											
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[illegible]

Calc

Electrical Store.ANL

MAX.	0.40	0.00	1001	0.00	0.00	1001			
	0.00	0.00	1001	0.00	0.00	1001	7.84	T	0.00 1221
	-0.40	0.00	6.98	1322	0.00	6.98	1322		
MIN.	0.00	0.00	6.98	1322	0.00	6.98	1322	19.13	T 6.98 1013

8003	0.00	MAX	0.40	1114	0.00	1322	0.00	1322	0.00	1322
	1.40	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	2.79	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.19	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	5.58	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.98	MAX	-0.26	1322	0.00	1322	0.00	1322	0.00	1322

MAX/MIN FORCE VALUES FOR MEMB 8003, AMONGST ALL SECT LOCATIONS

RAW/PLR FORCE VALUES FOR MEMB											
	FZ	DIST	LD	FZ	DIST	LD	FX	DIST	LD		
MAX.	0.40	0.00	1001	0.00	0.00	1001	6.40	T	6.98	1321	
				0.00	0.00	1001					
MIN.	-0.40	0.00	6.98	1322	0.00	6.98	1322	19.13	T	0.00	1013
				0.00	6.98	1322					

8004	0.00	MAX	0.40	1114	0.00	1322	0.00	1322	0.00	1322
	1.40	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	2.79	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	4.19	MAX	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	5.58	MIN	0.00	1322	0.00	1322	0.00	1322	0.00	1322
	6.98	MAX	-0.26	1322	0.00	1322	0.00	1322	0.00	1322

* STAAD SPACE

* STAAD SPACE

MAX/MIN FORCE VALUES FOR MEMB 8004, AMONGST ALL SECT LOCATIONS

MAX/MIN FORCE VALUES FOR MEMB				8004, AMONGST ALL SECT LOCATIONS					
FZ	FY	DIST	LD	MZ	DIST	LD	FX	DIST	LD
MAX.	0.40	0.00	1001	0.00	0.00	1001	5.28	T	6.98 1322
MIN.	-0.40	0.00	6.98 1114	0.00	6.98 1322	0.00	6.98 1322	23.52	T 0.00 1012

***** END OF FORCE ENVELOPE FROM INTERNAL STORAGE *****

176. LOAD LIST 2001 2002 2011 TO 2014

177. PRINT JOINT DISPLACEMENTS ALL

1 JOINT DISPLACE ALL

* STAAD SPACE

* STAAD SPACE

JOINT DISPLACEMENT (CM RADIANS)

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
102	2001	0.0188	-0.8212	-0.3274	0.0008	-0.0001	0.0030
	2002	0.0189	-0.8181	-0.2838	0.0007	-0.0001	0.0031
	2011	0.0186	-0.8038	-0.2504	0.0009	-0.0005	0.0031
	2012	0.0188	-0.8077	-0.2504	0.0011	-0.0002	0.0027
	2013	0.0149	-0.7302	-0.4279	0.0011	-0.0004	0.0025
	2014	0.0117	-0.6250	-0.1781	0.0004	-0.0016	0.0028
104	2001	0.0245	-0.9196	0.3460	0.0016	0.0014	0.0028

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2002	0.0208	0.7828	0.3348	0.0013	0.0024	0.0019
2011	0.0123	-0.2747	8.2441	0.0045	0.0012	0.0014
2012	0.0184	-0.1744	8.2544	0.0051	0.0013	0.0014
2013	0.0319	-1.2202	0.4134	0.0020	0.0017	0.0038
2014	0.0121	-0.3861	0.2817	0.0007	0.0011	0.0011
110	2001	0.0188	0.2761	-0.0009	0.0001	0.0031
	2002	0.0189	0.2582	-0.0008	0.0001	0.0031
	2011	0.0190	0.2544	-0.0007	0.0001	0.0031
	2012	0.0192	0.2544	-0.0007	0.0001	0.0031
	2013	0.0149	0.2544	-0.0007	0.0001	0.0031
	2014	0.0117	0.2544	-0.0007	0.0001	0.0031
502	2001	-0.0186	0.2544	-0.0007	0.0001	0.0031
	2002	-0.0188	0.2544	-0.0007	0.0001	0.0031
	2011	-0.0189	0.2544	-0.0007	0.0001	0.0031
	2012	-0.0189	0.2544	-0.0007	0.0001	0.0031
	2013	-0.0149	0.2544	-0.0007	0.0001	0.0031
	2014	-0.0117	0.2544	-0.0007	0.0001	0.0031
504	2001	-0.0245	-0.9196	0.3460	0.0016	0.0014
	2002	-0.0208	-0.7828	0.3348	0.0013	0.0014
	2011	-0.0223	-0.2747	8.2441	0.0045	0.0012
	2012	-0.0184	-0.1744	8.2544	0.0051	0.0013
	2013	-0.0319	-1.2202	0.4134	0.0020	0.0017
	2014	-0.0121	-0.3861	0.2817	0.0007	0.0011
108	2001	0.0242	-0.9119	0.3517	0.0016	0.0014
	2002	0.0206	-0.7790	0.3322	0.0013	0.0014
	2011	0.0165	-0.7790	0.3322	0.0013	0.0014
	2012	0.0117	-0.7790	0.3322	0.0013	0.0014
	2013	0.0116	-0.7790	0.3322	0.0013	0.0014
	2014	0.0116	-0.7790	0.3322	0.0013	0.0014
508	2001	-0.0242	-0.9119	0.3517	0.0016	0.0014
	2002	-0.0206	-0.7790	0.3322	0.0013	0.0014
	2011	-0.0165	-0.7790	0.3322	0.0013	0.0014
	2012	-0.0117	-0.7790	0.3322	0.0013	0.0014
	2013	-0.0117	-0.7790	0.3322	0.0013	0.0014
	2014	-0.0117	-0.7790	0.3322	0.0013	0.0014
510	2001	-0.0188	-0.8156	0.2582	0.0008	0.0031
	2002	-0.0189	-0.8156	0.2582	0.0008	0.0031
	2011	-0.0190	-0.8156	0.2582	0.0008	0.0031
	2012	-0.0192	-0.8156	0.2582	0.0008	0.0031
	2013	-0.0149	-0.8156	0.2582	0.0008	0.0031
	2014	-0.0117	-0.8156	0.2582	0.0008	0.0031

* STAAD SPACE

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--- PAGE NO. ---

JOINT DISPLACEMENT (CM RADIANS)

STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
201	2001	0.0000	0.0000	0.0000	-0.0014	-0.0003	-0.0005
	2002	0.0000	0.0000	0.0000	-0.0012	-0.0003	-0.0005
	2011	0.0000	0.0000	0.0000	-0.0012	-0.0003	-0.0005
	2012	0.0000	0.0000	0.0000	-0.0012	-0.0003	-0.0005
	2013	0.0000	0.0000	0.0000	-0.0012	-0.0003	-0.0005
	2014	0.0000	0.0000	0.0000	-0.0012	-0.0003	-0.0005
202	2001	0.0008	-0.0558	-0.2383	0.0008	0.0003	0.0010
	2002	0.0006	-0.0488	-0.2394	0.0007	0.0001	0.0010
	2011	0.0005	-0.0488	-0.2394	0.0007	0.0001	0.0010
	2012	0.0005	-0.0488	-0.2394	0.0007	0.0001	0.0010
	2013	0.0005	-0.0488	-0.2394	0.0007	0.0001	0.0010
	2014	0.0005	-0.0488	-0.2394	0.0007	0.0001	0.0010
203	2001	-0.0024	-0.0328	-0.1506	0.0014	0.0004	0.0004
	2002	-0.0024	-0.0328	-0.1506	0.0014	0.0004	0.0004
	2011	-0.0024	-0.0328	-0.1506	0.0014	0.0004	0.0004
	2012	-0.0024	-0.0328	-0.1506	0.0014	0.0004	0.0004
	2013	-0.0024	-0.0328	-0.1506	0.0014	0.0004	0.0004
	2014	-0.0024	-0.0328	-0.1506	0.0014	0.0004	0.0004
204	2001	-0.0329	-0.0941	-0.2509	0.0008	0.0003	0.0005
	2002	-0.0329	-0.0941	-0.2509	0.0008	0.0003	0.0005
	2011	-0.0329	-0.0941	-0.2509	0.0008	0.0003	0.0005
	2012	-0.0329	-0.0941	-0.2509	0.0008	0.0003	0.0005
	2013	-0.0329	-0.0941	-0.2509	0.0008	0.0003	0.0005
	2014	-0.0329	-0.0941	-0.2509	0.0008	0.0003	0.0005
205	2001	0.0063	-0.2231	-0.0713	0.0006	0.0001	0.0006
	2002	0.0063	-0.2231	-0.0713	0.0006	0.0001	0.0006
	2011	0.0063	-0.2231	-0.0713	0.0006	0.0001	0.0006
	2012	0.0063	-0.2231	-0.0713	0.0006	0.0001	0.0006
	2013	0.0063	-0.2231	-0.0713	0.0006	0.0001	0.0006
	2014	0.0063	-0.2231	-0.0713	0.0006	0.0001	0.0006
206	2001	0.0073	-0.3510	-0.1375	0.0006	0.0001	0.0006
	2002	0.0073	-0.3510	-0.1375	0.0006	0.0001	0.0006
	2011	0.0073	-0.3510	-0.1375	0.0006	0.0001	0.0006
	2012	0.0073	-0.3510	-0.1375	0.0006	0.0001	0.0006
	2013	0.0073	-0.3510	-0.1375	0.0006	0.0001	0.0006
	2014	0.0073	-0.3510	-0.1375	0.0006	0.0001	0.0006

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STAD SPACE	0.0000	0.0404	8.1328	Electrical	Store,AWL	0.0000
2011	0.0000	-0.1670	8.1539	0.0003	0.0000	0.0000
2012	0.0000	-0.3561	-0.0270	0.0001	0.0000	0.0000
2013	0.0000	-0.2652	-0.0270	0.0000	0.0000	0.0000
2014	0.0000	-0.2328	-0.0276	-0.0009	0.0000	0.0000
307	2001	0.0000	-0.1678	-0.0007	0.0000	0.0000
2002	0.0000	-0.0958	8.1241	0.0000	0.0000	0.0000
2011	0.0000	-0.0894	-0.0141	0.0071	0.0000	0.0000
2012	0.0000	-0.2496	8.1470	0.0014	0.0000	0.0000
2013	0.0000	-0.2776	-0.0279	-0.0010	0.0000	0.0000
2014	0.0000	-0.1313	-0.0289	-0.0005	0.0000	0.0000
308	2001	0.0000	-0.1440	0.0019	0.0000	0.0000
2002	0.0000	-0.1188	0.0087	0.0042	0.0000	0.0000
2011	0.0000	-0.3392	8.0334	0.0014	0.0000	0.0000
2012	0.0000	-0.4265	8.0822	0.0033	0.0000	0.0000
2013	0.0000	-0.1740	0.0064	-0.0016	0.0000	0.0000
2014	0.0000	-0.0634	-0.0133	-0.0007	0.0000	0.0000
309	2001	0.0000	-0.0380	0.0000	0.0000	0.0000
2002	0.0000	-0.0494	0.0048	-0.0014	0.0000	0.0000
2011	0.0000	-0.0494	7.7974	0.0071	0.0000	0.0000
2012	0.0000	-0.6566	7.7930	0.0061	0.0000	0.0000
2013	0.0000	-0.0704	0.1266	-0.0017	0.0000	0.0000
2014	0.0000	-0.0357	0.0416	0.0000	0.0000	0.0000
310	2001	0.0000	-0.0399	0.0078	0.0000	0.0000
2002	0.0000	-0.0346	0.1680	-0.0006	0.0000	0.0000
2011	0.0000	-0.7529	6.9396	0.0105	0.0000	0.0000
2012	0.0000	-0.7719	7.1363	0.0097	0.0000	0.0000
2013	0.0000	-0.0484	0.2424	-0.0010	0.0000	0.0000
2014	0.0000	-0.0254	0.0963	-0.0005	0.0000	0.0000
311	2001	0.0000	0.0000	0.0011	0.0000	0.0000
2002	0.0000	0.0000	0.0000	0.0009	0.0000	0.0000
2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2012	0.0000	0.0000	0.0000	0.0019	0.0000	0.0000
2013	0.0000	0.0000	0.0000	0.0212	0.0000	0.0000
2014	0.0000	0.0000	0.0000	0.0013	0.0000	0.0000
401	2001	0.0000	0.0000	0.0006	0.0000	0.0000
2002	0.0000	0.0000	0.0000	-0.0014	0.0003	0.0003
2011	0.0000	0.0000	0.0000	-0.0012	0.0003	0.0003
2012	0.0000	0.0000	0.0000	0.0137	0.0003	0.0003
2013	0.0000	0.0000	0.0000	0.0037	0.0003	0.0003
2014	0.0000	0.0000	0.0000	-0.0003	0.0001	-0.0001
402	2001	0.0000	0.0000	0.0008	0.0001	-0.0001
2002	0.0000	-0.0556	0.2843	0.0000	0.0000	-0.0010
2011	0.0000	-0.0488	6.2394	0.0007	0.0001	-0.0010
2012	0.0000	-0.6963	6.7390	0.0109	0.0005	-0.0010
2013	0.0000	0.6963	6.7000	0.0314	0.0002	-0.0007
2014	0.0000	0.6677	-0.3582	0.0011	0.0002	-0.0004
2015	0.0024	-0.0328	-0.1506	0.0004	0.0001	--

JOINT DISPLACEMENT (CM)		RADIALS		STRUCTURE TYPE = SPACE			
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
403	2001	0.0405	-0.0764	-0.1778	0.0016	-0.0018	-0.0005
	2002	0.0443	0.0668	-0.1458	0.0014	-0.0020	-0.0006
	2011	0.0447	0.0677	-0.5246	0.0078	-0.0021	-0.0005
	2012	0.0487	0.0585	-0.5309	0.0083	-0.0022	-0.0007
404	2013	0.0329	-0.0941	-0.2202	0.0020	-0.0017	-0.0005
	2014	0.0349	0.0428	-0.0972	0.0008	-0.0025	-0.0011
	4001	-0.0053	-0.1770	-0.0615	0.0016	-0.0014	-0.0012
	2002	-0.0044	0.3544	-0.0444	0.0014	-0.0013	-0.0010
405	2011	-0.0091	0.0315	7.9239	0.0045	-0.0012	-0.0009
	2012	-0.0087	0.1930	7.9672	0.0050	-0.0012	-0.0009
	2013	-0.0063	-0.2231	-0.0713	0.0037	-0.0017	-0.0016
	2014	-0.0041	-0.0912	-0.0411	0.0010	-0.0011	-0.0004
406	2001	-0.0817	-0.2735	-0.2111	0.0009	-0.0001	0.0021
	2002	-0.0781	-0.4026	-0.0656	0.0019	-0.0001	0.0019
	2011	-0.0768	-0.0908	8.0630	0.0023	-0.0001	0.0020
	2012	-0.0422	-0.3510	-0.0274	0.0013	-0.0002	0.0024
407	2013	-0.0773	-0.1375	-0.3256	0.0005	-0.0001	0.0022
	2014	-0.1084	-0.3595	-0.0257	0.0000	0.0000	0.0028
	4001	-0.1036	-0.3146	-0.0128	0.0000	0.0000	0.0028
	2002	-0.1036	-0.0816	8.0055	-0.0001	0.0000	0.0026
408	2011	-0.1035	-0.0816	8.0649	-0.0001	0.0000	0.0026
	2012	-0.0966	-0.1840	-0.0257	0.0000	0.0000	0.0033
	2013	-0.1229	-0.4547	-0.0257	0.0000	0.0000	0.0028
	2014	-0.1007	-0.1757	-0.0243	0.0000	0.0002	0.0022
409	2001	-0.0811	-0.2778	-0.0115	-0.0010	0.0001	0.0021
	2002	-0.0778	-0.2428	-0.0115	-0.0009	0.0001	0.0021
	2011	-0.0811	-0.1374	8.0006	0.0017	0.0001	0.0023
	2012	-0.0724	-0.2101	8.0621	0.0013	0.0001	0.0020
2013	-0.0915	-0.3513	-0.0239	-0.0013	0.0002	0.0024	

Page 24

Year	STAD SPACE	Electrical Store, MW	0.0033
2013	0.1229	-0.04547	-0.0033
2014	0.1067	-0.0257	-0.0033
2015	0.0811	-0.1758	-0.0023
207	0.0778	-0.0243	-0.0021
2002	0.0778	-0.5428	-0.0021
2003	0.0811	-0.0113	-0.0023
2011	0.0811	8.0006	-0.0001
2012	0.0724	-0.1374	-0.0001
2013	0.0724	8.0821	-0.0001
2014	0.0915	-0.3513	-0.0024
2015	0.0915	-0.0239	-0.0002
2016	0.0766	-0.0257	-0.0001
208	0.0052	-0.1758	-0.0011
2002	0.0044	-0.1538	-0.0011
2011	0.0050	-0.0190	-0.0010
2012	0.0046	7.9251	-0.0006
2013	0.0040	-0.3869	-0.0006
2014	0.0062	7.9584	-0.0004
2015	0.0062	-0.0205	-0.0015
2016	0.0040	-0.0097	-0.0004
2017	0.0040	-0.0899	-0.0011

JOINT DISPLACEMENT (CM) RADIAN(S) STRUCTURE TYPE = SPACE																
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN									
209	2001	-0.0406	-0.0728	0.1299	-0.0016	-0.0018	0.0005									
	2002	-0.0443	-0.0650	0.1219	-0.0014	-0.0020	0.0006									
	2003	-0.0529	-0.6294	7.5829	0.0070	-0.0070	0.0008									
	2011	-0.0561	-0.6467	7.6849	0.0066	-0.0024	0.0009									
	2013	-0.0329	-0.0805	0.1723	-0.0021	-0.0017	0.0005									
210	2001	-0.0549	-0.0393	0.0493	-0.0008	-0.0025	0.0011									
	2002	0.0006	-0.0513	0.2421	-0.0009	0.0001	0.0010									
	2003	0.0006	-0.0466	0.2182	-0.0008	0.0001	0.0010									
	2011	0.0005	-0.7474	6.8400	0.0107	0.0001	0.0011									
	2012	0.0004	-0.7397	6.9755	0.0104	0.0002	0.0011									
211	2001	-0.0027	-0.0652	0.3139	-0.0011	-0.0002	0.0007									
	2002	-0.0024	-0.0683	0.3083	-0.0003	-0.0002	0.0005									
	2003	0.0000	0.0000	0.0000	0.0000	0.0000	-0.0001									
	2011	0.0000	0.0000	0.0000	0.0000	0.0002	-0.0003									
	2012	0.0000	0.0000	0.0000	0.0000	0.0002	-0.0003									
301	2001	0.0000	0.0000	0.0000	0.0000	0.0001	-0.0005									
	2002	0.0000	0.0000	0.0000	0.0000	0.0001	-0.0005									
	2003	0.0000	0.0000	0.0000	0.0000	-0.0004	0.0004									
	2011	0.0000	0.0000	0.0000	-0.0011	0.0000	0.0000									
	2012	0.0000	0.0000	0.0000	-0.0009	0.0000	0.0000									
302	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2003	0.0000	0.0000	0.0000	-0.0014	0.0000	0.0000									
	2011	0.0000	-0.0026	-0.2303	-0.0007	0.0000	0.0000									
	2012	0.0000	-0.0359	7.0843	0.0006	0.0000	0.0000									
303	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
304	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
305	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
STAD SPACE	2001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2002	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2003	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2011	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
	2012	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000									
							-- PAGE NO. 38									

JOINT DISPLACEMENT (CM)		RADIAN(S)		STRUCTURE TYPE = SPACE			
JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
306	2001	0.0000	-0.3004	-0.0270	0.0001	0.0000	0.0000
2002		0.0000	-0.2404	-0.0335	0.0000	0.0000	0.0000

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ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
8002 ST UA90X90X6	PASS	BS-4.6 (T)	0.044	2013	6.98
8003 ST UA90X90X6	PASS	BS-4.6 (T)	0.044	2013	0.00
8004 ST UA90X90X6	PASS	BS-4.6 (T)	0.054	2012	0.00
19,60 T		0.00	0.00		0.00

***** END OF TABULATED RESULT OF DESIGN *****

182. STEEL TAKE OFF ALL
STAAD SPACE -- PAGE NO. 44

STEEL TAKE OFF

PROFILE	LENGTH(METE)	WEIGHT(KN)
ISCT: TAPERED	201	81.77
ISCT: TAPERED	3001	25.779
ST UA90X90X6		25.981
		27.92
		2.274
TOTAL =		53.134

***** END OF DATA FROM INTERNAL STORAGE *****

183. FINISH

***** END OF THE STAAD.PRO RUN *****

**** DATE= APR 28,2007 TIME= 15:20:42 ****

2 FOUNDATION DESIGN CHECK

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Calculation/Sketch

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Site	Main Depot Rashidiya - Building 24	Date	Apr '07	Job No.	068010
	Footing Summary		Designed by Rudy	Sheet No.	

Soil Bearing Pressure	=	150	kN/m ²
Soil Density	=	20	kN/m ³
Concrete Density	=	24	kN/m ³
Footing Depth below slab soffit	=	1	m
Column Stump Area	=	0.25	m ²

Individual Footing

Column	Column Reaction (ULS)	Soil Load (kN)	Footing Load (kN)	Total SLS Load (kN)	ULS Load (kN)	Footing size			Area of Footing (m ²)	Soil Pressure (kN/m ²)	Factor	Remarks
						L (m)	B (m)	H (m)				
201	85	34.2	28.224	133	172	1.400	1.400	0.600	1.96	68	2.21	OK

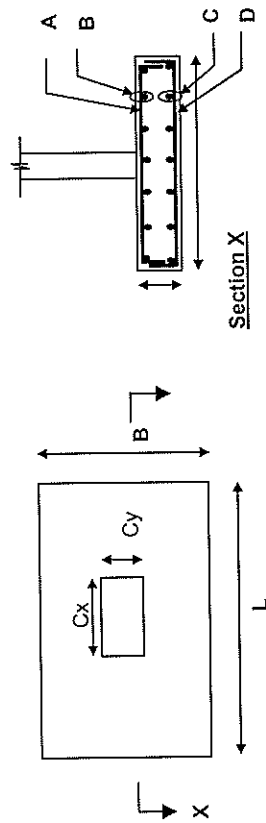
Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 24 Footing Detail	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Summary



Footing Type	L	B	H	Soil Bearing pressure (kPa)	A	Spacing (mm)	B	Spacing (mm)	C	Spacing (mm)	D	Spacing (mm)
F3	1400	1400	600	150	10T12	143	10T12	143	10T16	143	10T16	143

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 24 Footing F3 For soil bearing pressure = 150kPa	Date	Apr '07	Job No.	068010.01
		Designed by	Rudy	Sheet No.	

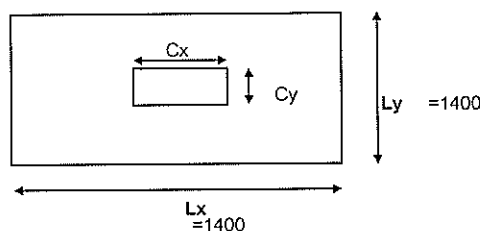
REINFORCED CONCRETE FOOTING DESIGN TO BS8110

1. Material Properties

Concrete grade	f_{cu}	=	35	N/mm ²
Steel grade	f_y	=	460	N/mm ²
Concrete density	γ_c	=	24	KN/m ³
Steel density	γ_s	=	78	KN/m ³

2. Design Data

Depth of Footing	h	=	600	mm
Width of Footing	L_y	=	1400	mm
Length of Footing	L_x	=	1400	mm
Width of Column	C_y	=	450	mm
Length of Column	C_x	=	450	mm
Concrete Ten. Cover	c	=	50	mm
Concrete Comp. Cover	c	=	50	mm
Tension bar dia. (x-dir)	D_x	=	16	mm
Tension bar dia. (y-dir)	D_y	=	16	mm
Comp. bar diameter (x-dir)	D_c	=	12	mm
Comp. bar diameter (y-dir)	D_c	=	12	mm
Effect.depth (x-dir)	d_x	=	542	mm
Effect.depth (y-dir)	d_y	=	526	mm
Area of footing	A	=	1.96	m ²



3. Loadings

Safe bearing pressure	p_b	=	150	KN/m ²
Load factor	f	=	1.5	
Total axial load (SLS)	N	=	294.00	kN
Eccentricity	e	=	0	mm

4. Analysis

Net pressure (SLS)	p_{max}	=	135.60	KN/m ²
Net pressure (ULS)	p_{max}	=	203.4	KN/m ²
Shear force (x-dir)	V_x	=	135.26	kN
Shear force (y-dir)	V_y	=	135.26	kN
Moment (x-dir)	M_x	=	32.12	kNm
Moment (y-dir)	M_y	=	32.12	kNm
Steel required (x-dir)	A_{sreq}	=	1092	mm ²
Steel provided (x-dir)	A_{sprov}	=	2011	mm ²
Steel required (y-dir)	A_{sreq}	=	1092	mm ²
Steel provided (y-dir)	A_{sprov}	=	2011	mm ²
Top bars required (x-dir)	A_{sreq}	=	1092	mm ²
Top bars provided (x-dir)	A_{sprov}	=	1131	mm ²
Top bars required (y-dir)	A_{sreq}	=	1092	mm ²
Top bars provided (y-dir)	A_{sprov}	=	1131	mm ²

Nos	ϕ	Spacing
10	16	143 x-dir
10	16	143 y-dir
10	12	143 x-dir
10	12	143 y-dir

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Main Depot Rashidiya - Building 24 Footling F3 For soil bearing pressure = 150kPa	Date	Apr '07	Job No.	068010.01
		Designed by	Rudy	Sheet No.	

5. Shear Check

Checking for punching shear:

Shear force at face $V_f = 395.44$ kN
Shear stress at face $v_f = 0.41$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (x-dir):

Shear force at face (x-dir) $V_c = 149.63$ kN
Shear stress at face (x-dir) $v_c = 0.20$ N/mm² O.K.

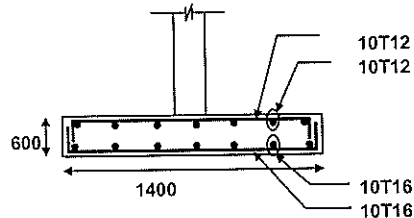
Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	542	0.00	0.00	0.45	0.91	No Shear Links
2d	1084	0.00	0.00	0.45	0.45	No Shear Links

Checking for shear (y-dir):

Shear force at face (y-dir) $V_c = 149.63$ kN
Shear stress at face (y-dir) $v_c = 0.20$ N/mm² O.K.

Distance from Support (mm)	av (mm)	V (kN)	v (N/mm ²)	v_c (N/mm ²)	Enhanced v_c (N/mm ²)	Remarks
d	526	0.00	0.00	0.46	0.92	No Shear Links
2d	1052	0.00	0.00	0.46	0.46	No Shear Links

6. Detailing



3 CONNECTIONS DESIGN

Document No.: **DM001-E-MAO-CDP-DR-DCC-371952**

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Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Column Splice Connection Design

From the STAAD Pro analysis

$F_{xc} =$	47	kN (compression)
$F_{xt} =$	22	kN (tension)
$F_y =$	20	kN
$M_z =$	75	kNm

Total Tension

$T = (n \cdot T_b / d) \cdot [(d-a) + (d-y_1-a) + (d-y_2-a) + \dots]$
where n = no of row of bolts and T_b is bolt tension capacity

Total Compression

$= 0.5a \cdot p_c \cdot b$
where b = stiff bearing length
 $= \text{web thickness} + 2 \cdot \text{plate thickness}$

Moment Capacity

$M_c = [n T_b / (d-a)] \cdot [(d-a)^2 + (d-a-y_1)^2 + (d-a-y_2)^2 + \dots]$

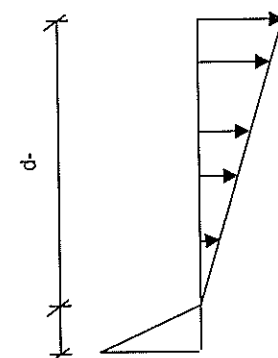
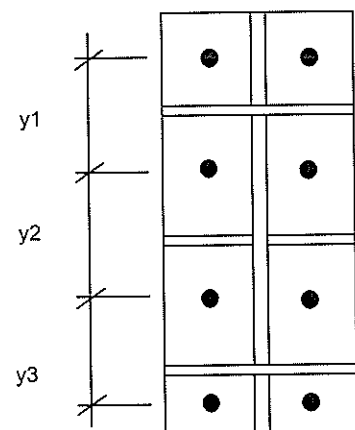
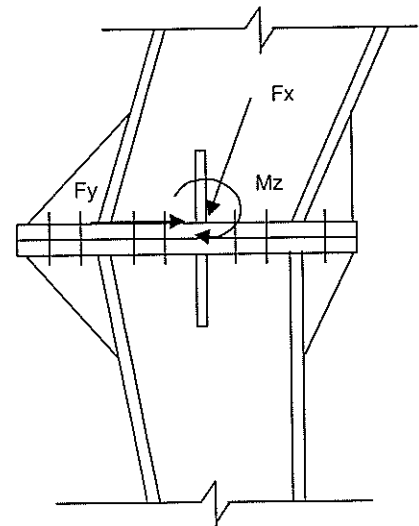
For Gr 8.8 bolt

Bolt size	=	20 mm
T_b	=	110 kN
V_b	=	91.9 kN
n	=	2
n_{total}	=	12
web, t_w	=	8 mm
plate, t_p	=	10 mm
b	=	28 mm

y_1	127	Trial a	=	84 mm
y_2	147	T	=	381.6 kN
y_3	127	C	=	381.6 kN
y_4	0	$T-C$	=	0.00 kN
y_5	0	M_c	=	95.8 kNm
y_6	0			
Depth, d	<u>400</u>			

$$\text{Tension force on the bolt, } T_1 = \frac{(M_z / M_c) \cdot T_b}{1.4} = 86 \text{ kN}$$

$$[T_1 + (F_{xt} / n_{\text{total}})] / T_b + F_y / (n_{\text{total}} \cdot V_b) = 0.82 < 1.4 \text{ Ok}$$



$$p_c = 325 \text{ N/mm}^2$$

Calculation/Sketch

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Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
			Designed by	Rudy	Sheet No.

Check the effective weld length on the stiffener plate to resist bolt's tension

Weld size, s = 6 mm (up to 15mm thick)
Weld strength, pw = 1.05 kN/mm (based on longitudinal capacity)
Weld length required = $(T1 + 0.5 \cdot F_{xt}) / pw$
= 92 mm
= 46 mm per face

Calculation/Sketch

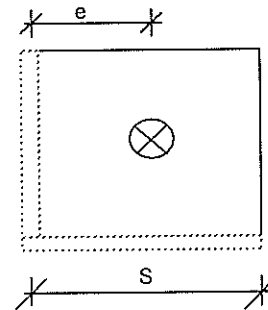
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Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
Designed by Rudy			Sheet No.		

From the plate analysis (refer attached) using
Prying Force, $T_o = 10 \text{ kN}$

Plate width, S_o (mm)	Ave. support bending moment, M_o (kNm)
75	0.020
100	0.027
125	0.035
150	0.040



Actual - Prying force, T	=	79 kN
Side, S	=	75 mm
Bending Moment, M	=	$(T/T_o) \cdot M_o$
	=	0.160 kNm

Bolt Diameter, ϕ	=	20 mm
Bolt distance, e	=	36 mm
Thickness of plate, t_p	=	10 mm
p_y allow	=	355 N/mm ²

Effective width of the plate resisting the bending moment, b_p	=	$\min (2e + \phi, S)$
	=	75 mm

Section of modulus of the plate, Z_p	=	$b_p \cdot t_p^2 / 6$
	=	1250 mm ²

M / Z_p	=	128 N/mm ²	Ok
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Calculation/Sketch

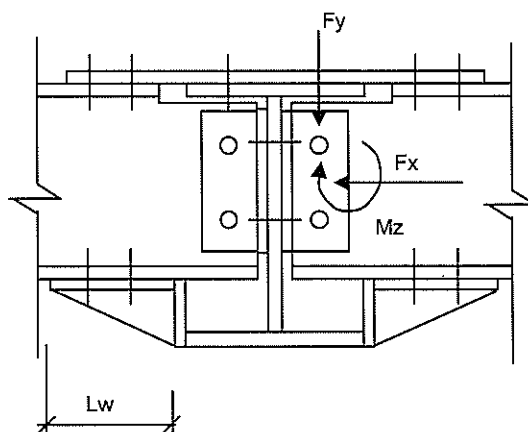
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Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

Secondary Beam Connection Design

Roof SDL	=	0.8 kN/m ²
Self weight	=	0.5 kN/m ²
WL	=	0.6 kN/m ² (downward)
WL	=	0.6 kN/m ² (downward)
	=	1.4 kN/m ² (upward)
wuls	-	1.4*DL+1.6LL = 2.78 kN/m ²
	-	1.2*(DL+LL+WL) = 3.00 kN/m ²
	-	1.4WL - DL = 0.66 kN/m ²
loading width, s	=	3.8 m



For internal span (simply supported span)

- top and bottom plate not required

max span, L = 8 m

Axial load for earthquake, Pa = (11.5+11.5)*(7.1) / 12 rows of secondary beams / tie beams
= 13.6 kN

Vmax = 0.5*wuls*L*s
Downward forces = 45.6 kN
Upward forces = 10.0 kN

Bolt diameter, ϕ = 16 mm
Shear capacity, Vb = 58.9 kN
Bolt tensile capacity, Tb = 70.7 kN

Check shear on bolt connected to secondary beam's web

No of bolt, n = 2
Vmax / n = 22.8 kN
Pa / n = 6.8 kN
Resultant shear, V = 23.8 kN Ok

Check tensile capacity of bolt connected to main beam's web

Eccentricity of load, e = 60 mm
Moment due to eccentricity, Me = Vmax*e
= 2.736 kNm
Spacing between bolt, db = 150 mm
Tensile force on the bolt, T = Me/db
= 18.2 kN Ok

Calculation/Sketch

Meinhardt Infrastructure Pte Ltd
Consulting Engineers, Planners & Managers



Site	Dubai Depot Electrical Store	Date	Apr '07	Job No.	068010
		Designed by	Rudy	Sheet No.	

For the cantilever end

- Assume shear due to bending is resisted by top and bottom bolt
- Assume vertical and horizontal shear is resisted by bolt on the web

Span length, L	=	4 m
Mmax	=	$0.5 \cdot wuls \cdot L^2 \cdot s$
Downward	=	91.2 kNm
Upward	=	20.1 kNm

Vmax	=	$wuls \cdot L \cdot s$
Downward	=	45.6 kN
Upward	=	10.0 kN

Check bolts on the flanges

Beam depth - flange thickness, Df	=	292 mm
Mmax bending moment (incl. moment due to ecc), Me	=	93.9 kNm

Shear on the flange, Vf	=	Me / Df
	=	321.7 kN

No of bolt provided	=	4
Shear per bolt	=	80.4 kN
Bolt diameter, ϕ	=	20 mm
Shear capacity, Vb	=	91.9 kN

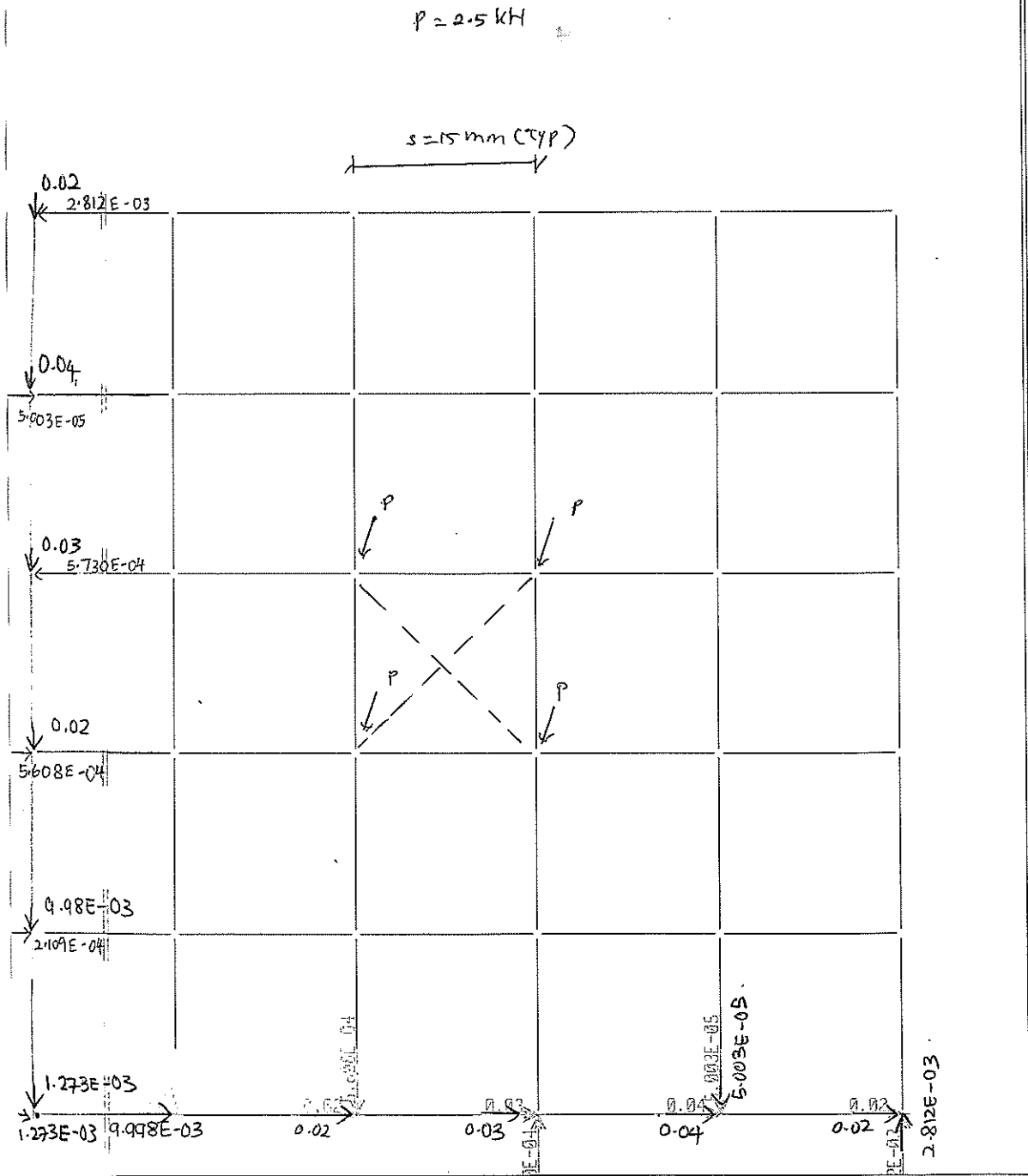
Check shear on the web

- for the secondary beam web

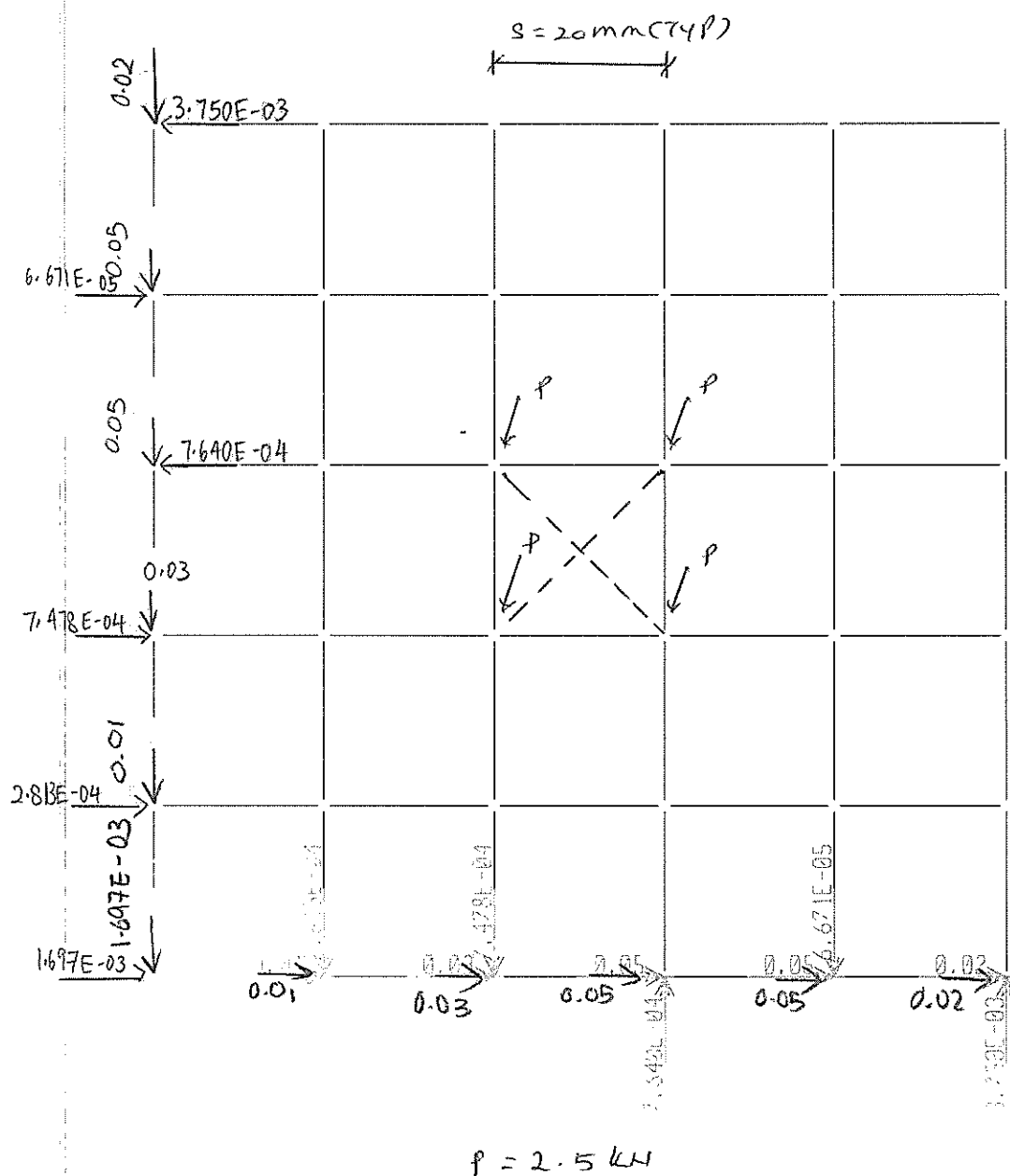
No of bolt provided	=	2
Shear per bolt	=	22.8 kN
Bolt diameter, ϕ	=	16 mm
Shear capacity, Vb	=	58.9 kN

- for the main beam web

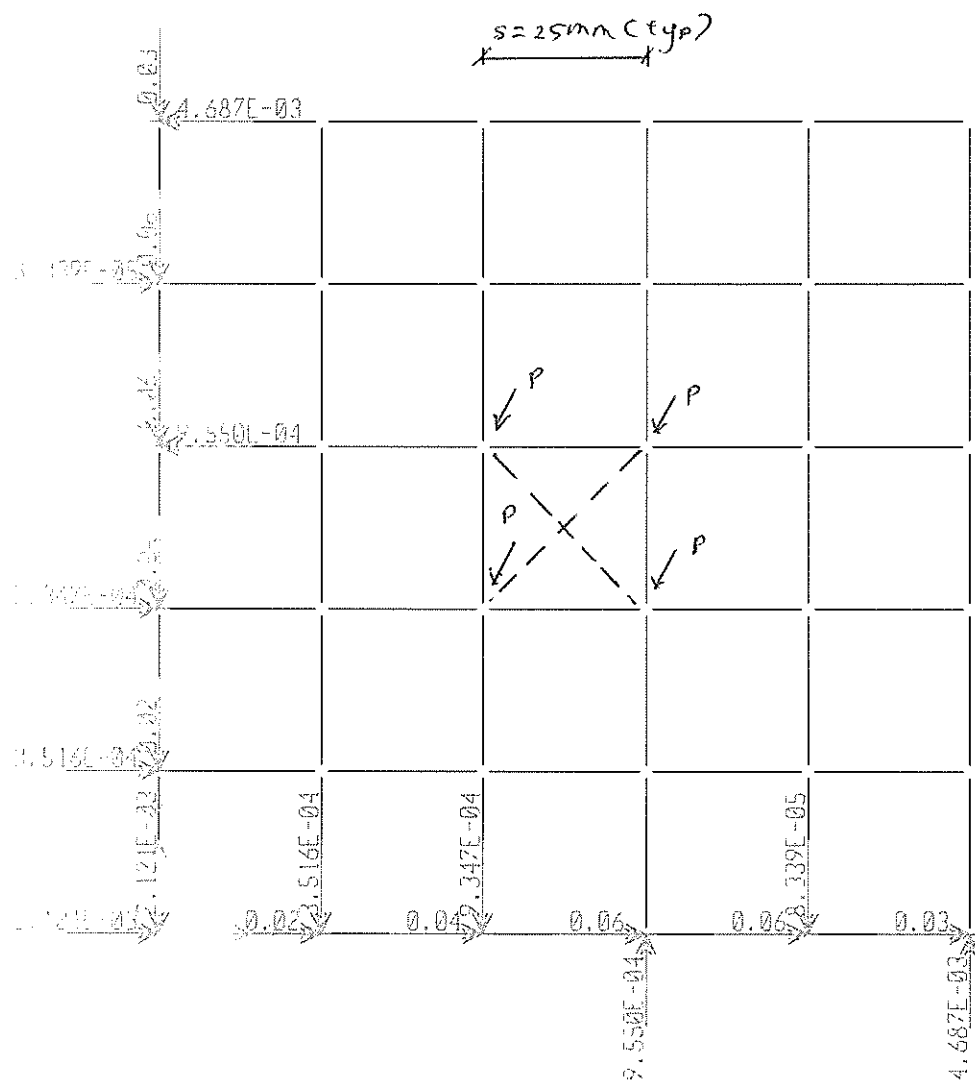
No of bolt provided	=	2
Distance between bolt	=	150 mm
Tensile load per bolt	=	18.2 kN
Bolt diameter, ϕ	=	16 mm
Tensile capacity, Vb	=	70.7 kN



Bolt Prying Action on Splice Plate
(Plate Width = 75 mm)

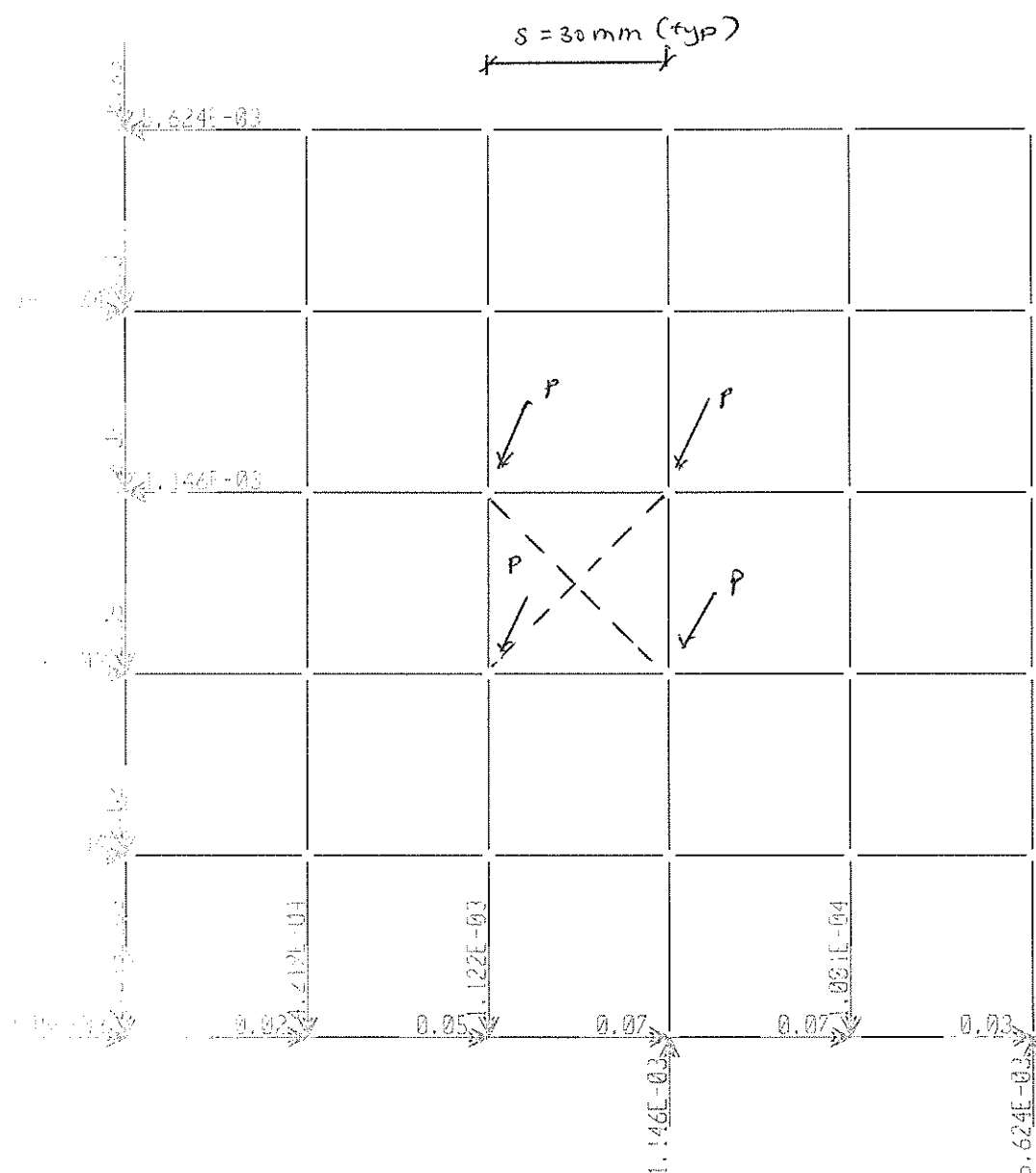


BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 100mm)



$$P = 2.5 \text{ kN}$$

BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 125mm)



$$P = 2.5 \text{ kN}$$

BOLT PRYING ACTION ON SPLICE PLATE
(PLATE WIDTH = 150mm)

